

Final Project Technical Report: Mixed-Signal Integrated Circuit Design and Verification

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1. Introduction and System Overview

This report seeks to outline the project workflow of a mixed-signal integrated circuit that implements a multi-mode switchable functionality. It is to meet the requirements set by the EE3300 curriculum at Iowa State University and its policies for final projects. This workflow includes the design, verification, and layout of a complete silicon level device in Cadence Virtuoso, using the GPDK045 CMOS process that was used in prior lab work.

A crucial operating condition set by the project specifications is the power supply that is provided for our integrated circuit. We are given 2.5V VDD and -2.5V VSS, which is crucial for a CMOS process such as this where both positive and negative DC signals as well as AC signals will be used. Our process actually ended up going with $\pm 1.8\text{V}$, as we had to use 2V logic transistors, and we received approval from Yash to do so. This accessibility grants the designer some freedom with scaling input signals without having to worry about clipping.

To handle the 4 different modes of the proposed device, a standard digital control scheme is implemented. Namely, a two-bit control signal is used as input from the user to control the active operating mode via a 2 to 4 bit decoder. This report will denote these input signals as A0 and A1, and the 4 modes as C0-C3. The operations of these four modes are outlined in the table below.

Mode Control (A0A1)	System Operating Mode	Primary Component Utilization
00	Programmable Inverting Amplifier	Op-Amp, Digital Potentiometer (Feedback)
01	Programmable Non-Inverting Amplifier	Op-Amp, Digital Potentiometer (Feedback)
10	16-Tap Digital Potentiometer	Resistor String, Transmission Gate Array
11	4-Bit Buffered Output String DAC	Op-Amp (Buffer), Resistor String, 1V Ref

First, when the mode control is set to $A0A1 = 00$, the device functions as a programmable inverting amplifier. In this state, the input analog signal is buffered and amplified based on the C0-C3 4-bit input, which controls gain. Second, when the mode control is set to $A0A1 = 01$, the integrated circuit operates as a programmable non-inverting amplifier. The architecture mirrors the inverting mode but routes the incoming signal to the non-inverting terminal of the core operational amplifier. In both active amplifier modes, the discrete gain steps are achieved by leveraging the third functionality of this circuit, the digital potentiometer, which is physically switched into the feedback loop of the operational amplifier.

Third, a digital state of $A0A1 = 10$, allows the device to act as a standalone 16-tap programmable potentiometer. This mode requires a highly linear physical taper, providing a strict impedance of 500 Ohms between any two adjacent taps across the resistor string. The bidirectional nature of the transmission gates utilized in this specific design permits the application of variable voltages on both the upper (V_H) and lower (V_L) terminals, bounded only by the supply rails. Finally, when the mode is set to $A0A1 = 11$, the system transitions into a 4-bit buffered Digital-to-Analog Converter. In this configuration, the digital potentiometer is re-tasked to serve as the precision resistor string, dividing a defined 1V reference voltage relative to ground. Concurrently, the internal operational amplifier is reconfigured into a unity-gain buffer to completely prevent output loading effects and ensure maximum analog signal integrity.

Below you can find a table of given specifications for this circuit, and the resulting target specifications for our operational modes.

Performance Metric	Target Specification	Design Tolerance / Notes
Non-Inverting Amplifier Gain	16 (Maximum)	Scalable via 16 discrete steps
Inverting Amplifier Gain	15 (Maximum)	Scalable via 16 discrete steps
Potentiometer Taps	16 Taps	Governed by 4 bit input control
Tap Impedance	500 Ohms	Linear taper layout implementation
DAC Levels	0 - 0.9375 V (with 1V reference)	Applied to potentiometer string

Aiming to closely match these target specifications required careful design decisions and frequent layout troubleshooting, particularly that of analog components within the dense GPDK045 process. The following sections will document the schematic formulation, the test

bench verification process, the layout generation, and the further optimizations.

2. Sub Circuit Level Design and Schematic Verification

Due to the multi leveled nature of this project, our design process took a bottom up/hierarchical approach. This required us to plan out the device and brainstorm what bottom level components that would be our starting point, devices that are made up of non-repeating transistor and passive component structures.

2.1 CMOS Transmission Gate

Given that this is a mixed signal project, there will be interaction between digital and analog signals. More specifically digital signals will be decoded and used as switches for analog signals to be re-routed and amplified. The CMOS Transmission Gate is a common and often necessary component that does just this, allowing an AC signal through its terminals due to its bidirectional capability.

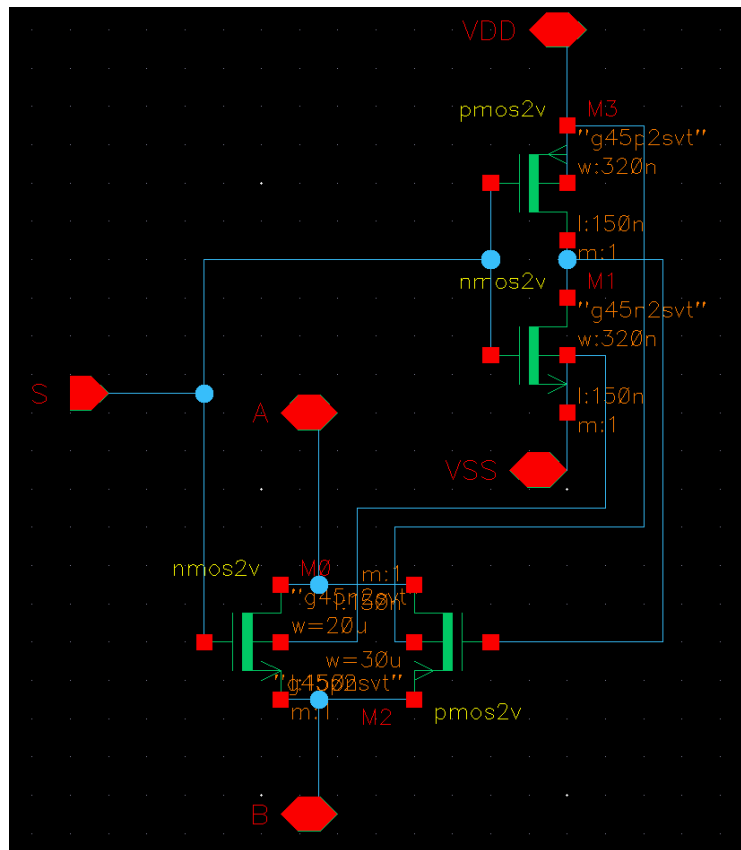
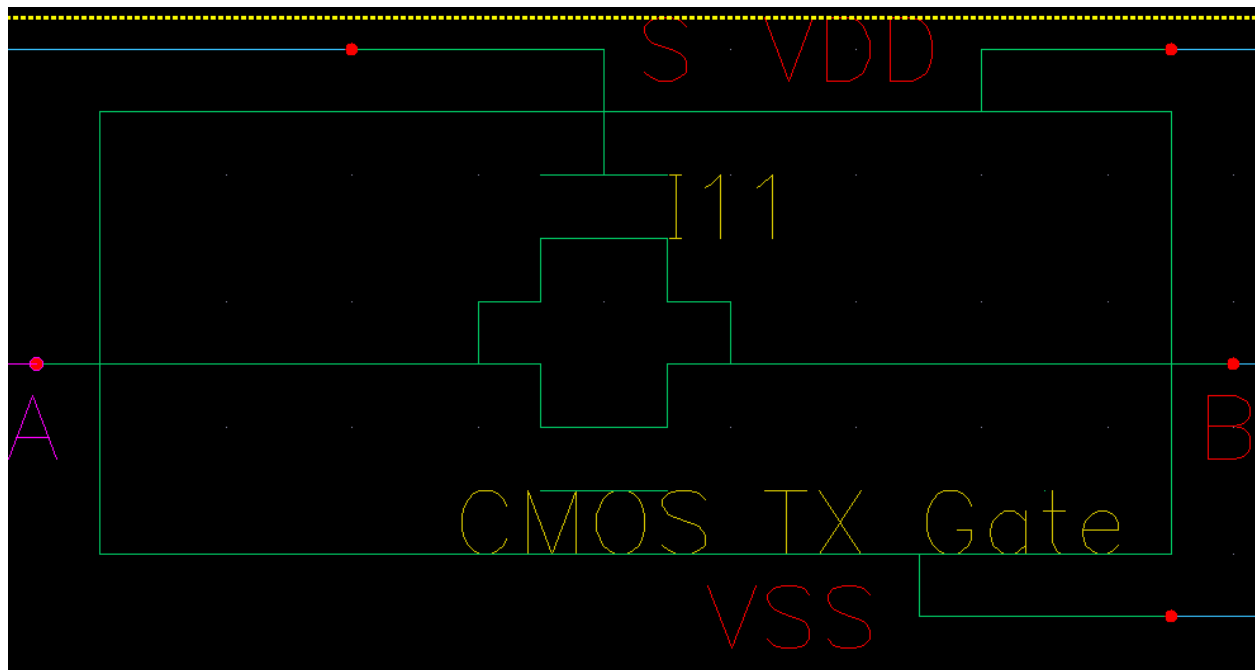


Figure 1: CMOS Transmission Gate Schematic

To ensure that the series resistance of these transmission gates is relatively negligible, we had to size them up substantially compared to the logic transistors surrounding them. This was mostly a

trial-and-error process, and we wanted to find a sweet spot where the size would not pose difficulties during layout, but the operational integrity was not compromised by their size. We found that an NMOS width of 20um against a length of 150nm was sufficient, which led us to size the PMOS three times as large, resulting in a width of 60um against a length of the same 150nm. After sizing these, we created a symbol to be used in larger schematics, which can be found below.



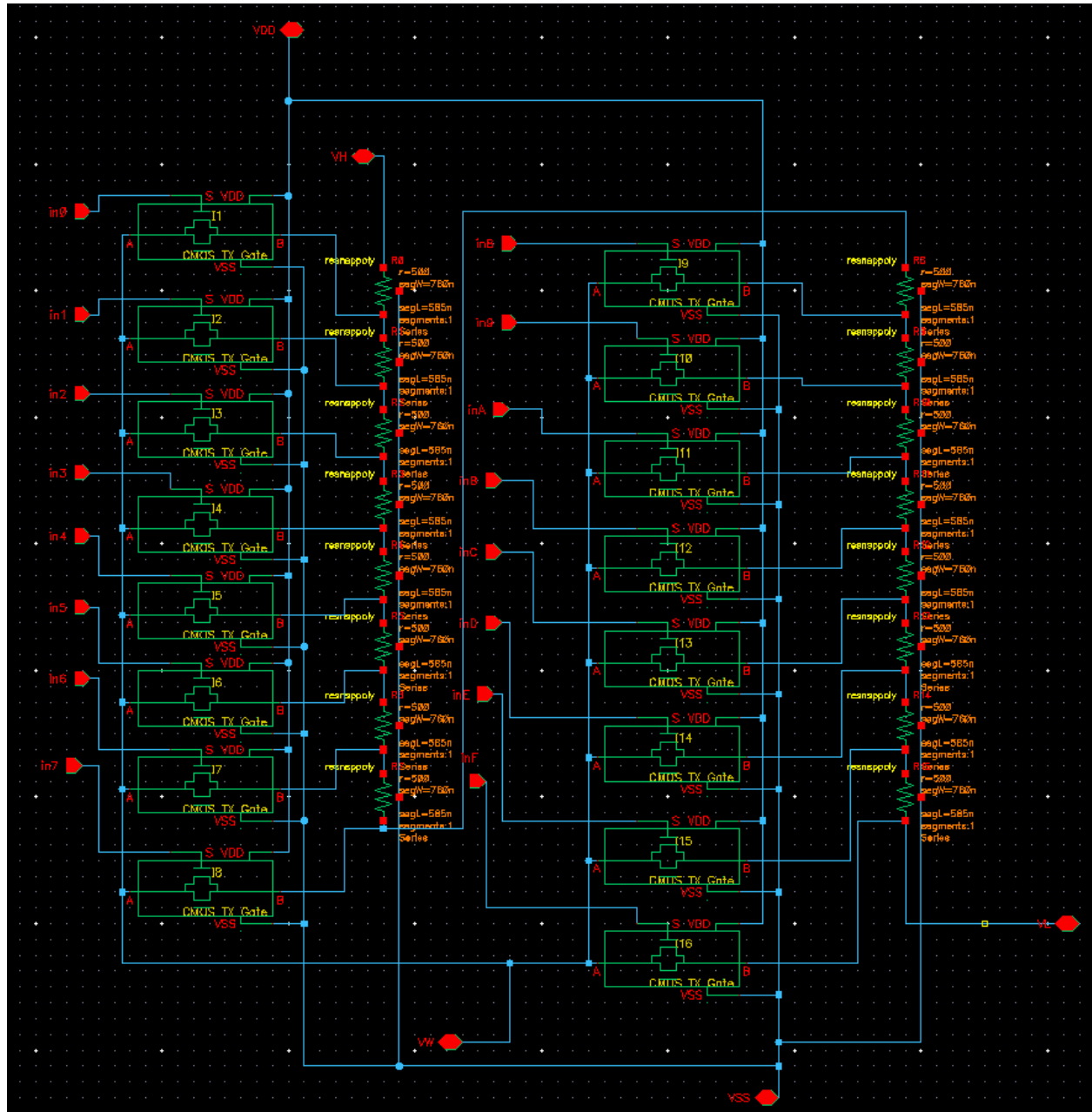
CMOS Transmission Gate Symbol

2.2 Digital Potentiometer

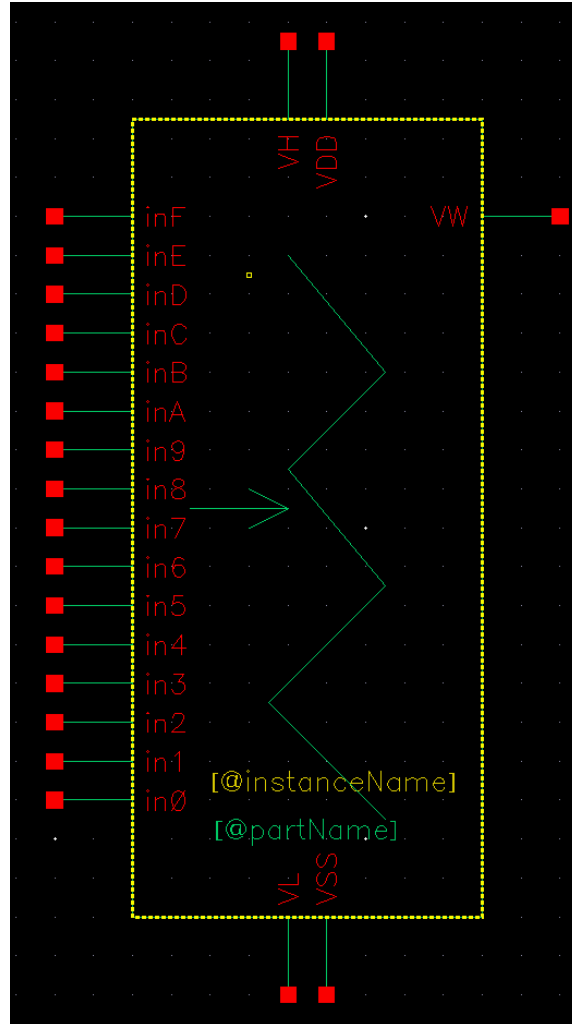
Every single mode of operation that our device works in requires a digitally-controlled potentiometer. More specifically, a potentiometer with 16 unique taps for the wiper, with the whole resistor networking totaling 8k ohms in 500-ohm increments. The easiest way that we identified to design this potentiometer was to place 16 500-ohm resistors in series, with a CMOS transmission gate attached to the node in between each of them, providing 16 unique access points. From there, the other transmission terminal of each of the transmission gates would be tied together, connected to the wiper terminal of the potentiometer device. Each of these transmission gates was controlled by the output of a 4-to-16 decoder, allowing 4-bit control of these 16 wiper positions.

With 16 wiper positions, this potentiometer can be used in each of our four modes of operation. In the case of the amplifier configurations, this potentiometer is used in the negative feedback network, providing a configurable resistance between the V_H and V_W (Wiper) terminals of

the potentiometer, thus providing an avenue for controlling the gain of the amplifiers. In the case of the potentiometer mode, the potentiometer can be used however the user desires: a voltage divider configuration with a grounded V_L terminal, or just as a variable resistor between V_H and the wiper. In the case of the DAC, the potentiometer is used in a voltage divider configuration to achieve variable output voltages.



Digital Potentiometer Schematic



Digital Potentiometer Symbol

2.3 Digital Control Logic and Decoders

To effectively convert user inputs into physical switch activation, we developed two digital decoders. A 4-to-16 decoder is necessary to control the location of the wiper in our digital potentiometer. More specifically, it controls the active connection to the resistor ladder that forms the backbone of our amplification and DAC. The 2-to-4 decoder is used similarly to convert an external two bit switch input into a toggle for the 4 modes of the device. For this logic, we naturally used a Hardware Description Language (HDL), specifically Verilog, due to its convenience and documented process for transferring into GPDK045 ready circuits. We drafted the code in the behavioral hierarchy due to the fact that a decoder is a lower level component that is easier to explicitly write each outcome for.

```

1  module projDecoder(out16, out4, in4, in2);
2      input [1:0] in2;
3      input [3:0] in4;
4      output [3:0] out4;
5      output [15:0] out16;
6
7      assign out4[0] = (~in2[1]) & (~in2[0]);
8      assign out4[1] = (~in2[1]) & (in2[0]);
9      assign out4[2] = (in2[1]) & (~in2[0]);
10     assign out4[3] = (in2[1]) & (in2[0]);
11
12     assign out16[0] = (~in4[3]) & (~in4[2]) & (~in4[1]) & (~in4[0]);
13     assign out16[1] = (~in4[3]) & (~in4[2]) & (~in4[1]) & (in4[0]);
14     assign out16[2] = (~in4[3]) & (~in4[2]) & (in4[1]) & (~in4[0]);
15     assign out16[3] = (~in4[3]) & (~in4[2]) & (in4[1]) & (in4[0]);
16     assign out16[4] = (~in4[3]) & (in4[2]) & (~in4[1]) & (~in4[0]);
17     assign out16[5] = (~in4[3]) & (in4[2]) & (~in4[1]) & (in4[0]);
18     assign out16[6] = (~in4[3]) & (in4[2]) & in4[1] & (~in4[0]);
19     assign out16[7] = (~in4[3]) & (in4[2]) & in4[1] & in4[0];
20     assign out16[8] = (in4[3]) & (~in4[2]) & (~in4[1]) & (~in4[0]);
21     assign out16[9] = (in4[3]) & (~in4[2]) & (~in4[1]) & in4[0];
22     assign out16[10] = (in4[3]) & (~in4[2]) & in4[1] & (~in4[0]);
23     assign out16[11] = (in4[3]) & (~in4[2]) & in4[1] & in4[0];
24     assign out16[12] = (in4[3]) & (in4[2]) & (~in4[1]) & (~in4[0]);
25     assign out16[13] = (in4[3]) & (in4[2]) & (~in4[1]) & in4[0];
26     assign out16[14] = (in4[3]) & (in4[2]) & in4[1] & (~in4[0]);
27     assign out16[15] = (in4[3]) & (in4[2]) & in4[1] & in4[0];
28
29     endmodule
30

```

projDecoder.v

Before synthesizing this code into logic blocks, it is important to verify that there are no mistakes in any assignment. This is done in QuestaSim, triggering each combination of inputs and viewing the results in a simulation, comparing those results to the ideal, expected behavior.

At first, we weren't sure if we should create separate modules for these decoders, so we made them separately at first. From there we decided it would be easier and more spatially-efficient to deal with a single module containing both devices. After cleaning up these misconfigurations, we had a working module, which can be seen by the clean diagonal cascade showing the expected performance of a decoder.

```

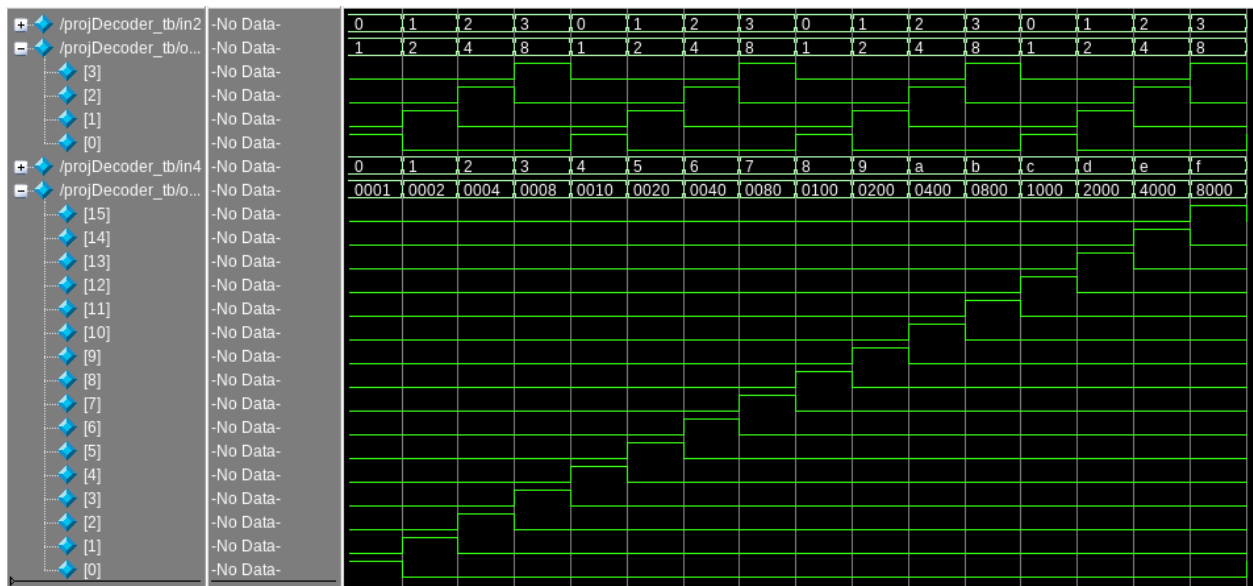
timescale 1ns/1ps
module projDecoder_tb();
    reg [1:0] in2;
    wire [3:0] out4;
    reg [3:0] in4;
    wire [15:0] out16;

    projDecoder dut(.out4(out4), .in2(in2), .out16(out16), .in4(in4));

    initial in2 = 2'b00;
    initial in4 = 4'b0000;
    always #5 in2 = in2 + 1;
    always #5 in4 = in4 + 1;
endmodule

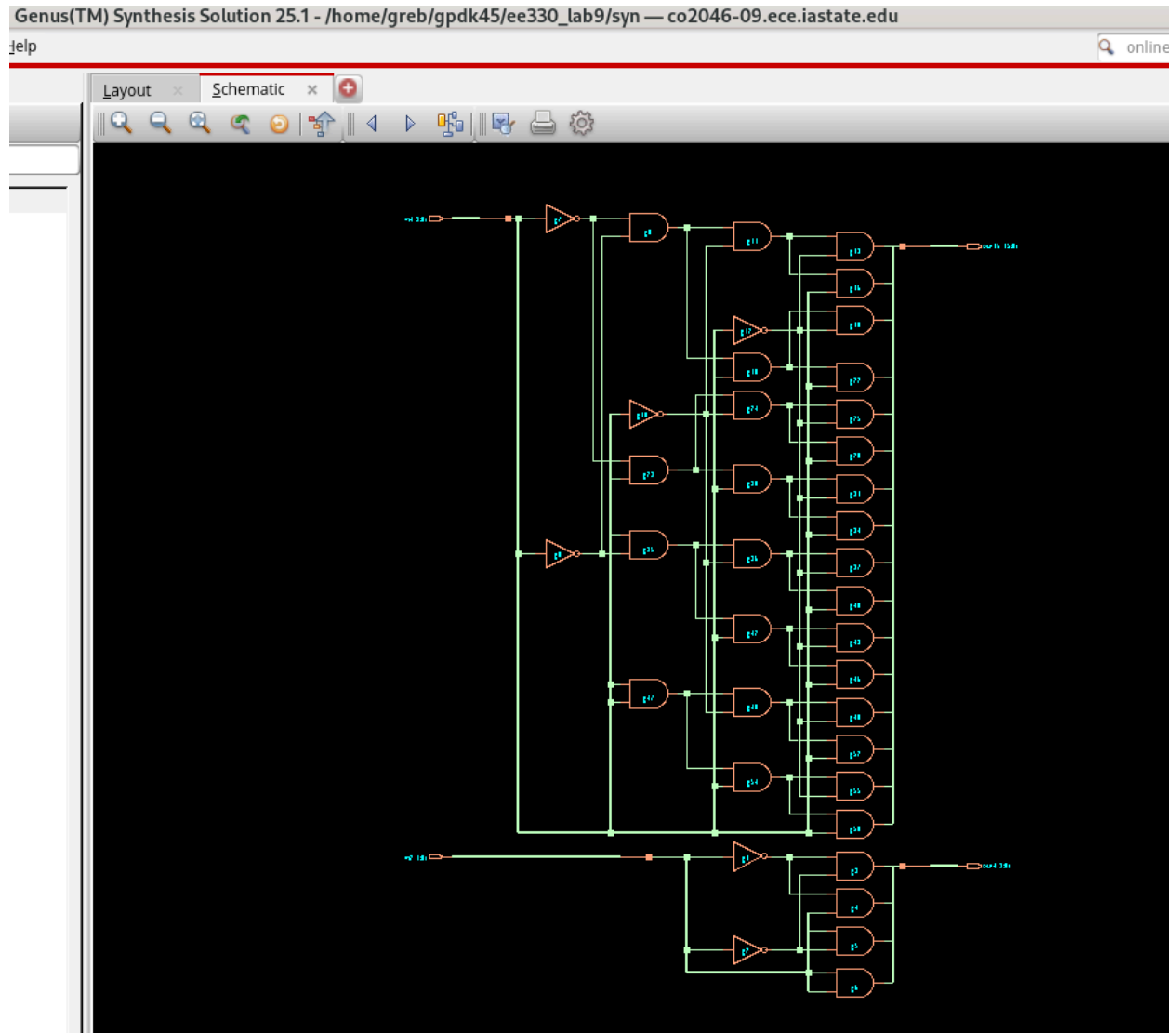
```

projDecoder Testbench Code



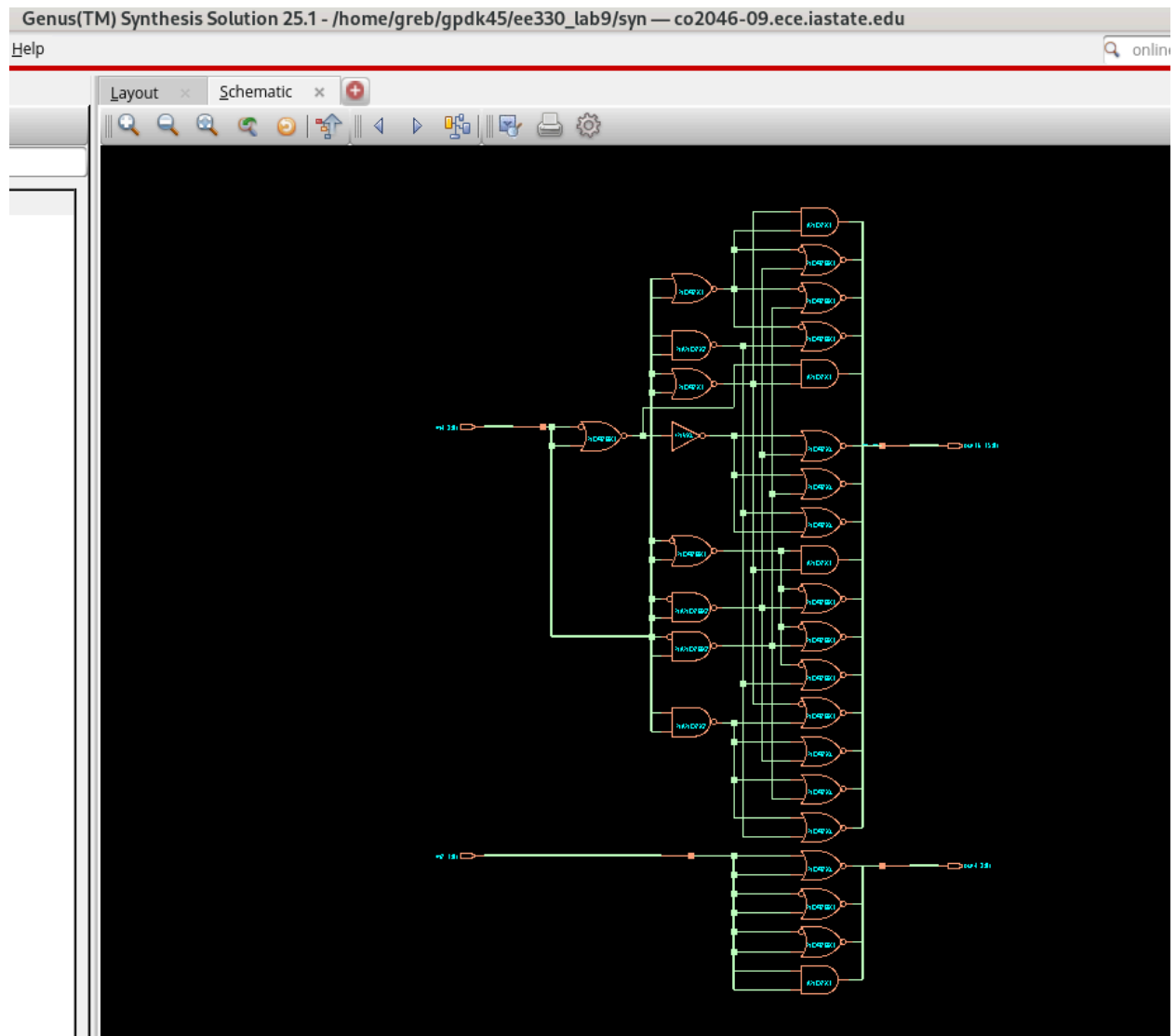
Testbench Results for projDecoder

Once the testbench was written to test each input value, we ran the testbench and observed the results. Fortunately, the first attempt was successful. With the code verified, we translated the module into primitive logical blocks within the Cadence Genus Synthesis tool, yielding a raw standard cell mapping.



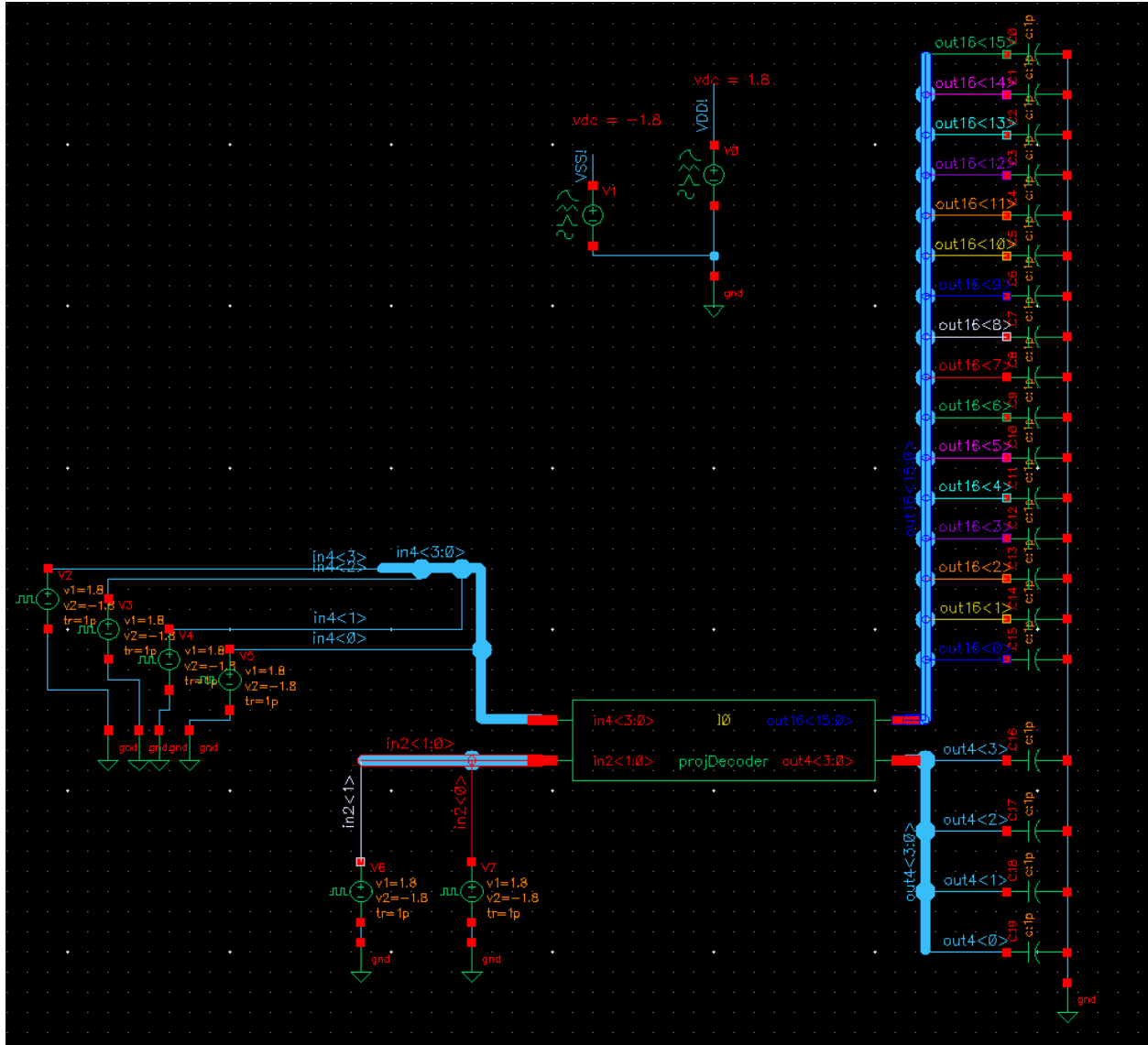
projDecoder in Genus

As the course has taught us to do in prior lab work, we then applied advanced optimization to minimize the gate count and compress the digital logic into its optimal solution, minimizing transistor counts. Given that these decoders are highly independent in each function assignment, it can be seen that the optimizer only saves about 6 or 7 gates worth of space. That being said, that is still about 24 transistors. In an expensive fabrication process, this step is essential and would make an even larger difference if we had more complicated logic circuits.



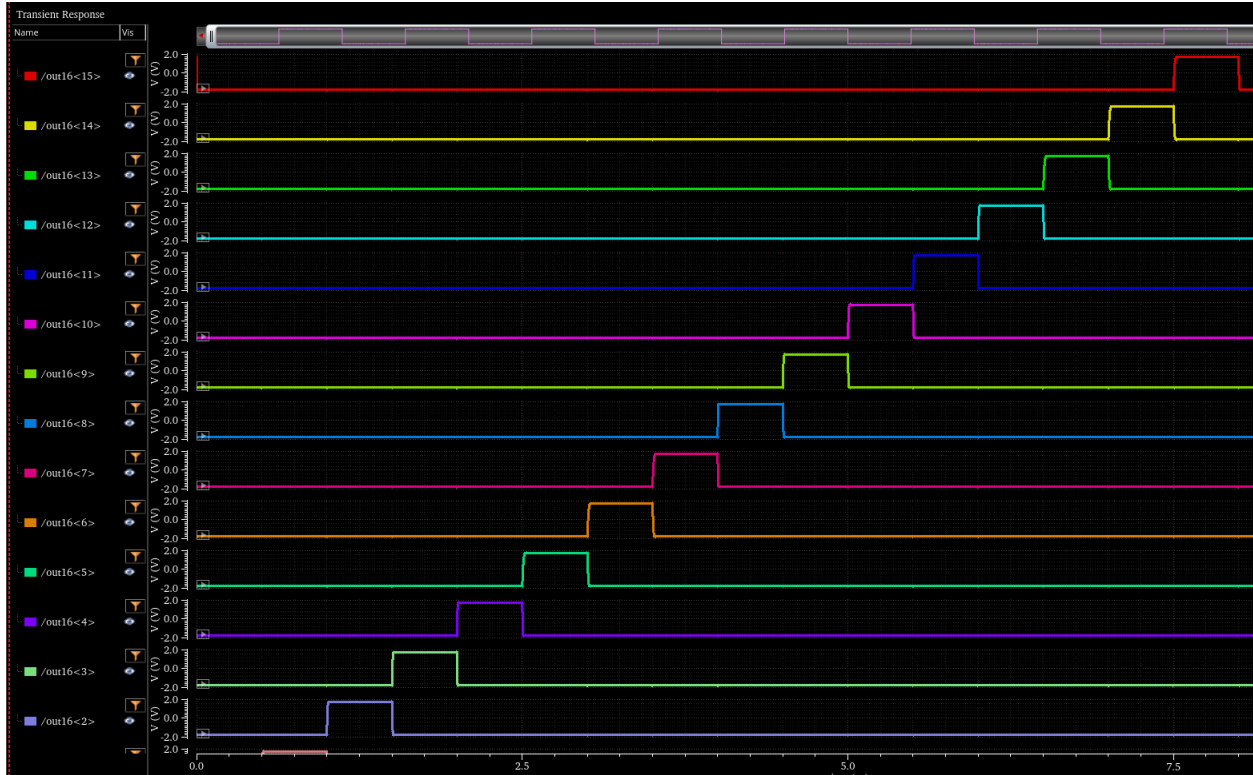
projDecoder Optimized in Genus

To verify the synthesized digital blocks, we established a test bench before proceeding with integration. First, the decoders were formed into a unified top-level symbol, entitled projDecoder. Setting 1.8V boolean inputs with decreasing switching frequencies at increasingly significant bits like we have in many other test benches for prior labs, and monitoring each decoder output we can compare the logic to the base code.



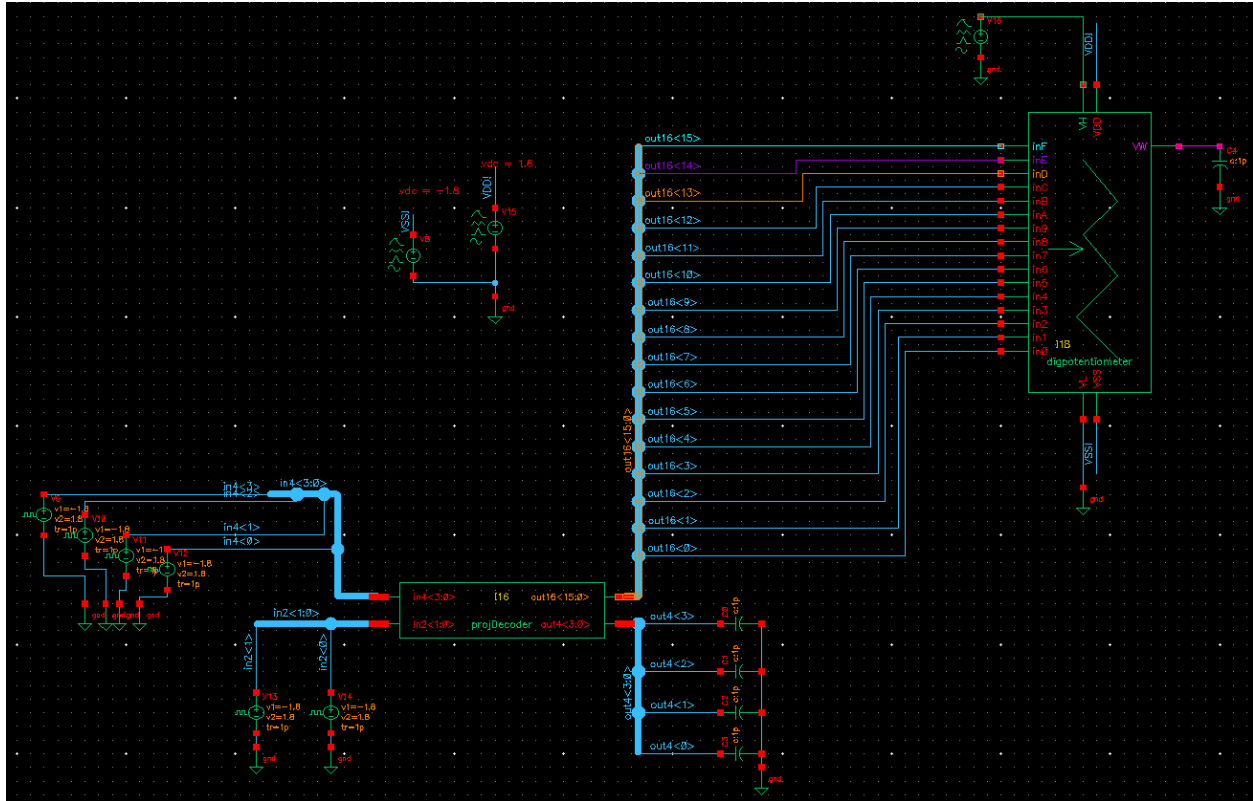
projDecoder Testbench in Cadence

Our transient waveforms confirmed the same sequential cascading of the decoder's logic levels, with relatively sharp transitions between the 1.8V high and -1.8V low states. This meant that we were ready to proceed to integrating this block in the larger functions of the whole circuit.



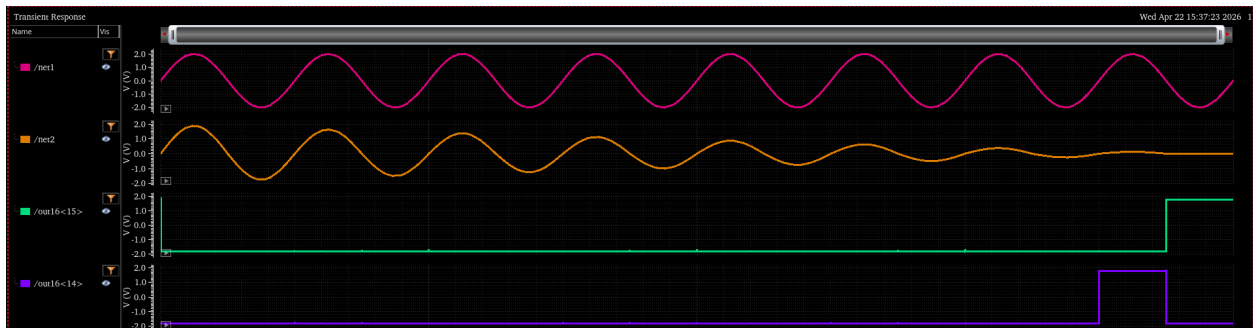
projDecoder Testbench Output

With the low-level digital verification complete, we wired the projDecoder to the digital potentiometer, attaching each of the 16 outputs of the 4-to-16 part of the module to the 16 transmission gates in the digital potentiometer. This represented the first mixed-signal interaction within our project architecture.



Decoder with Potentiometer Testbench

The resulting transient waveforms proved that our logic signals possessed sufficient electrical drive strength to manipulate the gates of the switches without detrimental loading adding in noise. The test bench was focused on extracting a scaled AC voltage based on the ascending digital input. As the 4 bit switches, the location of the wiper changes, acting as the variable voltage divider, aka potentiometer, that we had planned. It can be seen that every half-cycle of the sinusoidal input/output the amplitude lowers, demonstrating the potentiometer/voltage divider behavior as the controlled input changes.

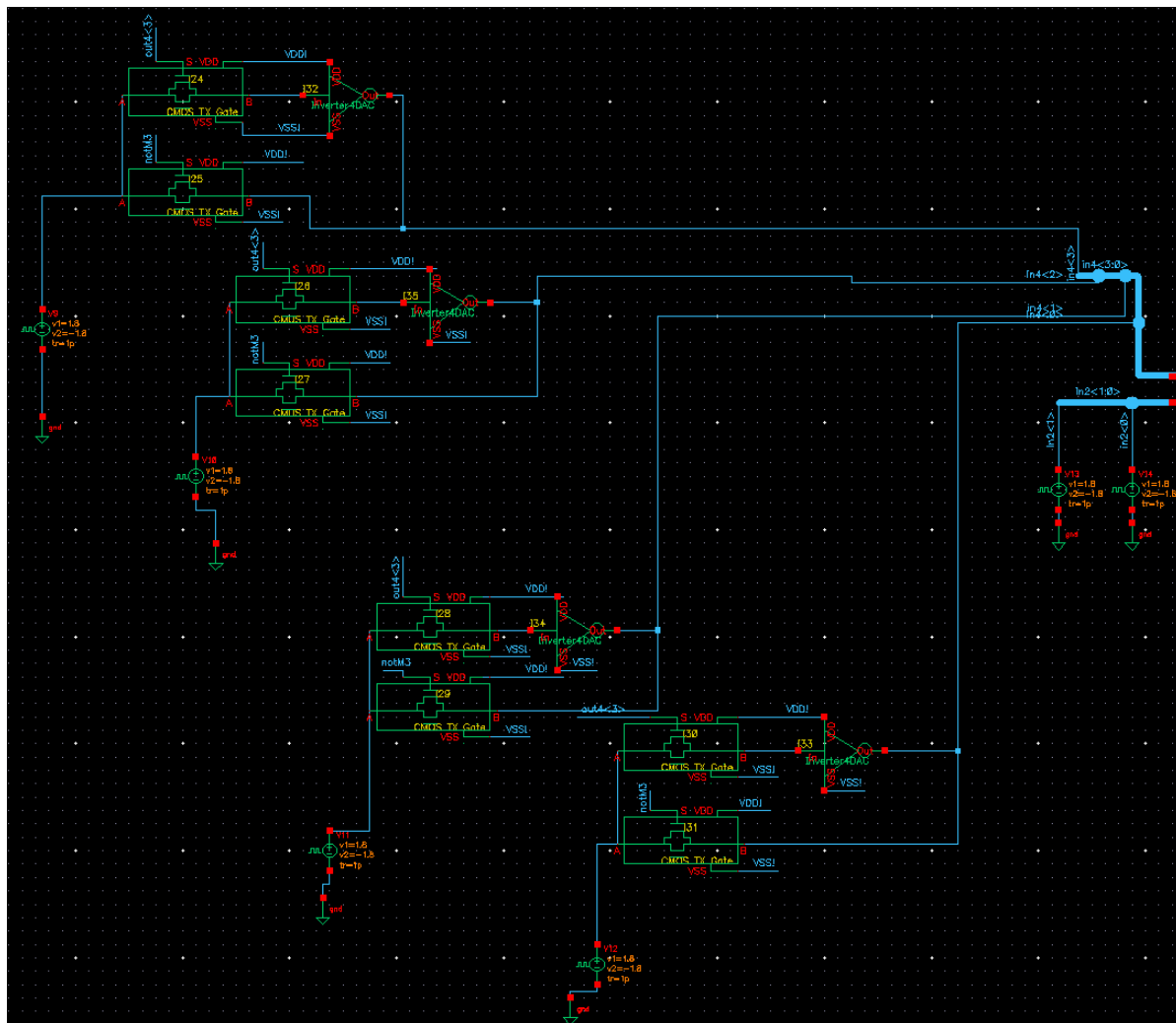


Decoder with Potentiometer Testbench Results

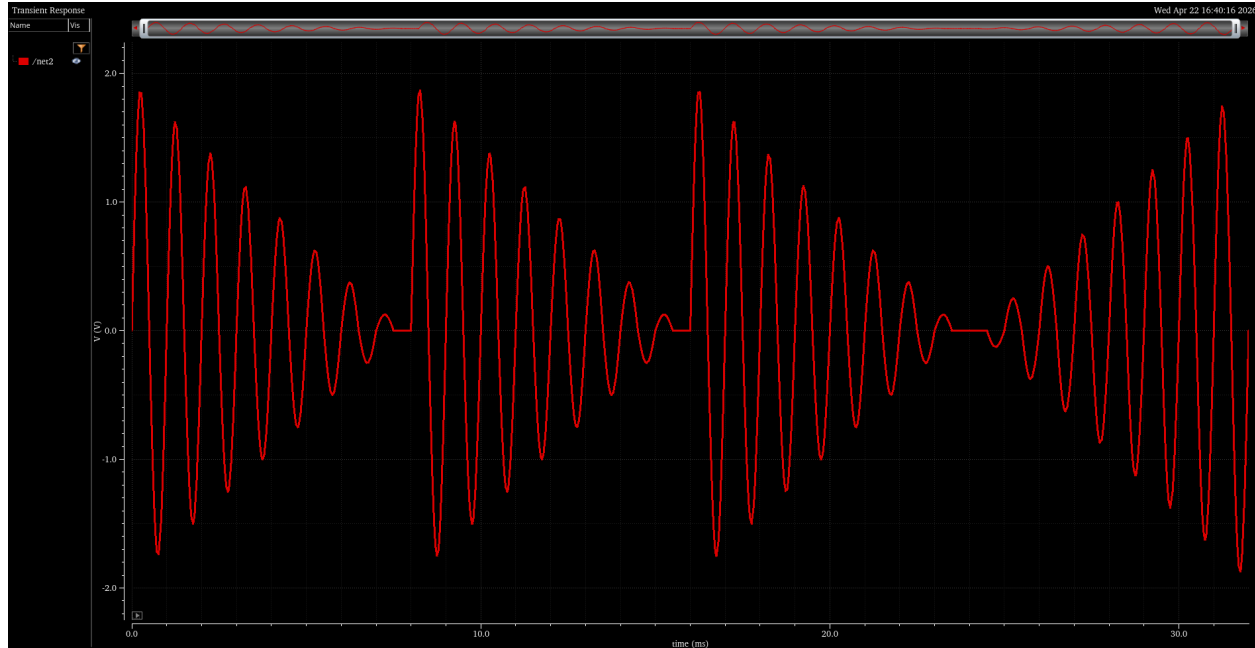
After verifying the functionality of both the decoder and the potentiometer working in tandem with the decoder, we identified that the 4-bit input to the decoder needed to behave differently based on the mode of operation: for mode 3 (DAC) the inputs needed to be inverted.

As seen above in the potentiometer testbench, the wiper voltage amplitude decreased as the 4-bit control input grew in size, which corresponded to the wiper moving further down the resistor ladder, leading to a lower wiper output. The DAC operates on the same principle as that, but with a 1V reference voltage provided at the V_H terminal. The difference with the DAC is that the wiper voltage must *increase* as the 4-bit input increases.

To allow the DAC to function properly, we wired transmission gates at each of the four-bit inputs that would invert the input to the decoder if the device is operating in mode 3 (DAC). This meant that a four-bit HIGH input would correspond to the highest DAC output, rather than the lowest, which would be the case if the inputs were not inverted. This control circuit can be seen below.



Potentiometer/DAC Control Circuit



Potentiometer/DAC Control Testbench Results

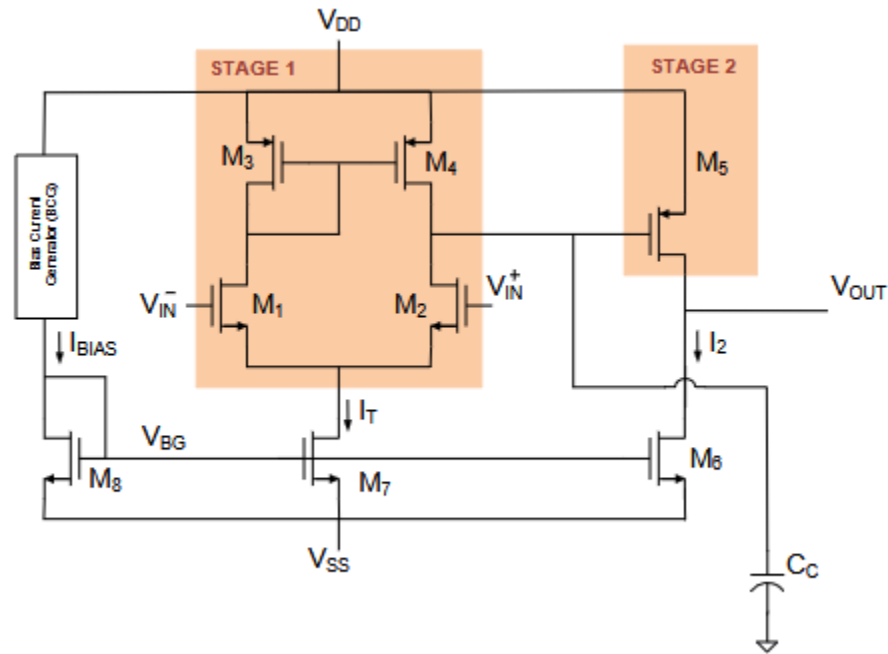
The testbench results above demonstrate this change in functionality described in the previous paragraph. For the first three modes, the output decreases in amplitude as time elapses and the 4-bit input grows. In the last mode (DAC), the output is at its lowest point when the four-bit gain control input is at 0000, and it reaches its maximum at the end, when the input is 1111: functionality that exhibits the exact opposite behavior of the previous three modes, as desired.

2.4 Operational Amplifier

Another bottom level component that is necessary for this circuit is an active analog device that can provide inverting and non inverting signal amplification and buffering for the DAC output. From our studies in EE2300 and EE3300, we know that we need an operational amplifier that can be fed by both input terminals. As for the specific architecture of the device, the team decided to go with the 7T Schematic design due to the convenience of the tools that our professors provided us to aid in the sizing of the device. The 7T architecture is also shown to be quite stable for DC and AC amplification, as outlined in “Externally Compensated Op Amps” by Randy Geiger in 2018.

To briefly analyze Geiger’s design, it includes a current biasing network that sets the current going through both stages of this device through the current mirror established between M6, M7, and M8. It is important to note that M8 isn’t necessarily required in this architecture, but its diode connected configuration makes setting the bias current much easier for the engineer. Stage 1 establishes a symmetrical current mirror with M3 and M4 so that the positive and negative inputs terminals on the gates of M1 and M2 are exactly balanced. Stage 2 is the primary driving

stage, providing the high gain of the op amp when used with stage 1. The shunt capacitor is used to change the circuit's frequency response, resulting in a more stable design.



Prof. Geiger's Designed 7T Op Amp Schematic (Geiger, 2018)

Using the sizing tool provided, we entered the design parameters of our GPDK045 process, referencing the GPDK Reference Manual from Princeton University. The model parameters were taken from our previous work in lab 8, where extraction of these parameters using the square law model was performed. We ensured to verify our old results with our TA and the reference documentation for the device.

Model Parameters

$\mu_n C_{OX}$	2.52E-04	AV^{*-2}
$\mu_p C_{OX}$	8.40E-05	AV^{*-2}
V_{Tn}	0.38	V
V_{Tp}	-0.38	V
λ	0.04635	V^{-1}
AV_{Tn0}	0.025	$V/\mu m$
AV_{Tp0}	0.025	$V/\mu m$

Design Parameters

V_{DD}	1.8	V
V_{SS}	-1.8	V
$A1$	400	μm^2
$A3$	400	μm^2
L_{min}	0.6	μm
LSF	4	

Updated April 18., 2018

By entering the areas of M1 and M3, lengths of M1,M2,M3 and M4 these areas and the W/L ratios. All other lengths will be the product of a reasonable value of the LSF is 4.

Data is entered in the blue-shaded regions. All other cells in this spreadsheet

Design Region

Performance Region

Design

|VEB|

Bias Currents (mA)

Gate Area

AV

VIN-CM

VOUT-Swing

Power

Vos(mV)

Active

M1

M3

M5

M6

IT

I2

IBIAS

M1

M3

(dB)

Min

Max

Min

Max

MW

3 sigma

Area (μm^2)

1

1

1

1

1

1

1

1

200

200

53.3580105

0.58

0.8

-0.8

0.8

10.8

10.6

1051.4

2

0.5

0.5

0.5

0.5

1

1

1

100

100

53.3580105

0.08

1.3

-1.3

1.3

10.8

7.5

800

3

4

5

6

7

8

9

10

$$\frac{W}{L} = \frac{2|I_{DQ}|}{\mu C_{OX} V_{GS}}$$

$$\sigma_{V_{OS}} = \sqrt{2 \left[\frac{A^2_{VTOn}}{A_1} + \frac{1}{A_3} \frac{V_{EB1}}{|V_{EB3}|} A^2_{VTOP} \right]}$$

Externally Compensated Op Amp Sizing Tool (Pt 1.)

The tool then calculated the device geometries and their requisite performance.

Device Geometries																							
W/L								Width (in μm)								Length (in μm)							
M1	M2	M3	M4	M5	M6	M7	M8	W1	W2	W3	W4	W5	W6	W7	W8	L1	L2	L3	L4	L5	L6	L7	L8
4.0	4.0	11.9	11.9	23.8	7.9	7.9	4.0	28.2	28.2	48.8	48.8	57.143	19.048	19.048	9.524	7.1	7.1	4.1	4.1	2.4	2.4	2.4	2.4
15.9	15.9	47.6	47.6	95.2	31.7	31.7	15.9	79.7	79.7	138.0	138.0	228.57	76.19	76.19	38.1	5.0	5.0	2.9	2.9	2.4	2.4	2.4	2.4

Externally Compensated Op Amp Sizing Tool (Pt 2.)

Note: With these images of the sizing tool is that they are not the exact same as the sizes we ended up using in Cadence. The sizes used in cadence are faithful to the official extracted parameters, while the image uses extracted parameters from our prior lab results.

To ensure maximum output swing, these transistors are frequently sized in pairs. This creates a symmetrical structure that avoids current imbalances within the circuit. The 7T architecture also requires a compensation capacitor at the gate of M5 (denoted M0 in the following figure). To our understanding, it is to create a dominant pole at a lower frequency to avoid potential nonideal

oscillations, but that is a bit out of the scope of this project. There was also a biasing current that had to be set between VDD and a central loading node. This was originally done by setting a resistor and voltage source in parallel. A quick schematic implementation of this design was made for preliminary testing.

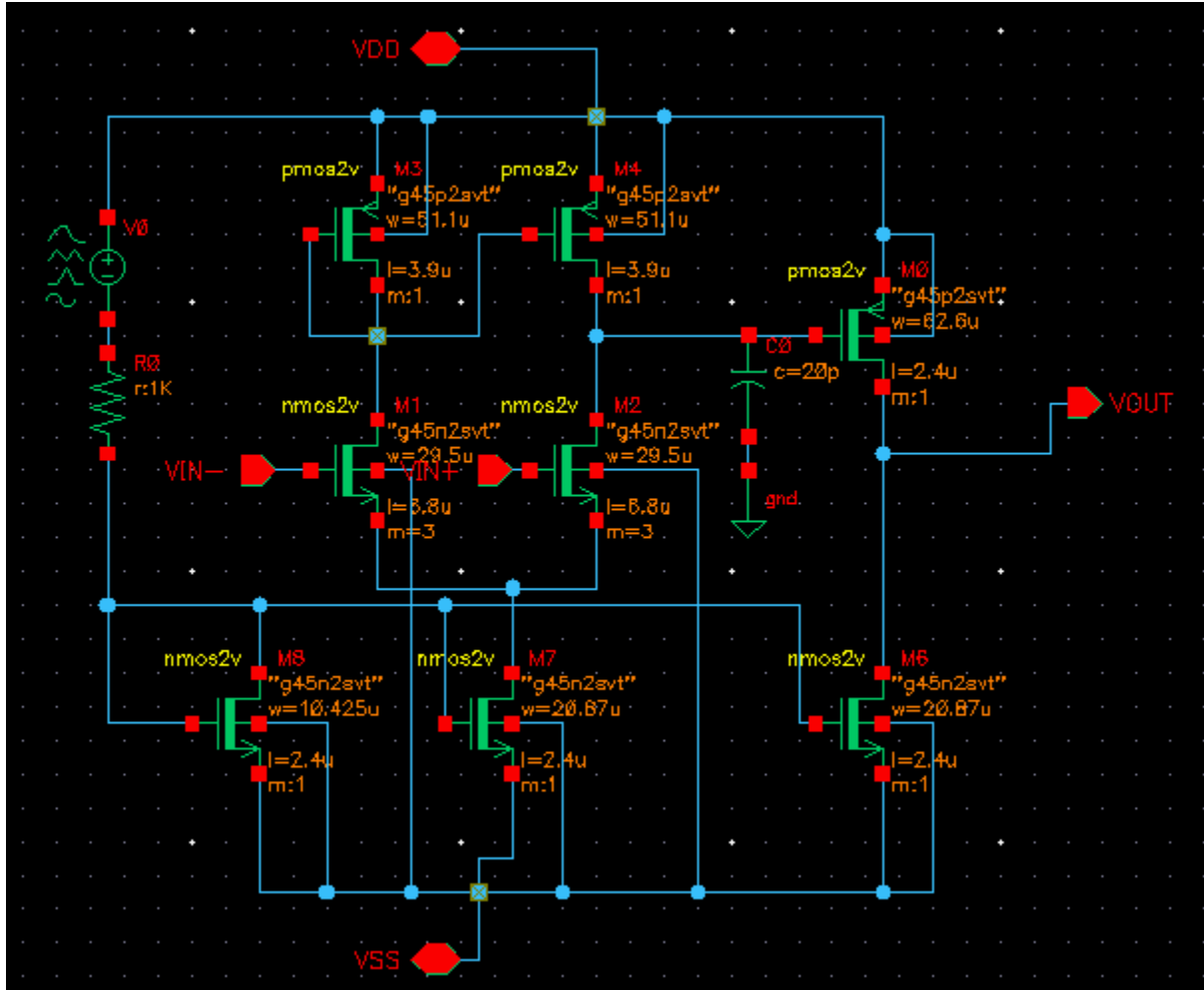
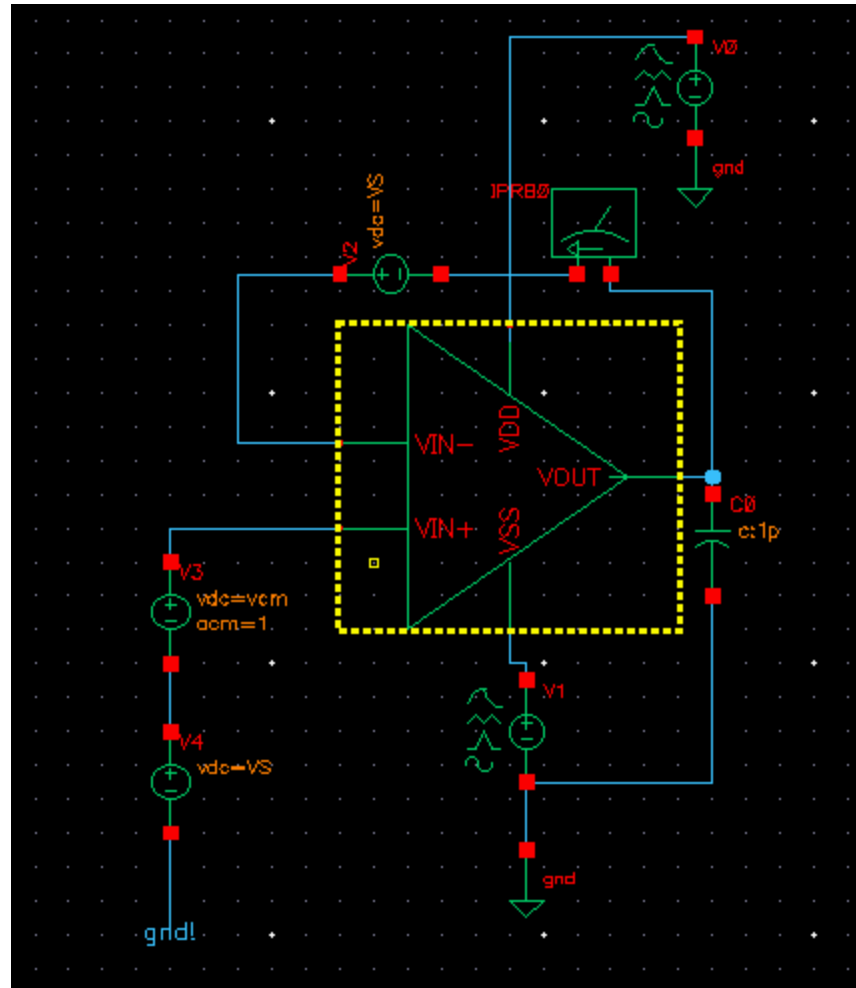


Figure 19: Initial 7T Op Amp Schematic

The next step is to verify that each of the transistors in the op-amp are operating in saturation and perform stability analysis on the design. This is done by supplying VDD and VSS to the rails, and supplying a variable V3 AC signal so that we can sweep frequency. There is also a variable DC voltage (VS) that is applied to both input terminals. An 1pF output capacitor to ground is used as standard practice. This can represent the following capacitive loading of the circuit once integrated into a larger system. Lastly, a stability probe (IPRB) is a Cadence tool that is added to measure open-loop gain and phase margin without affecting the behavior of the device. With this knowledge, a stability test bench is developed.



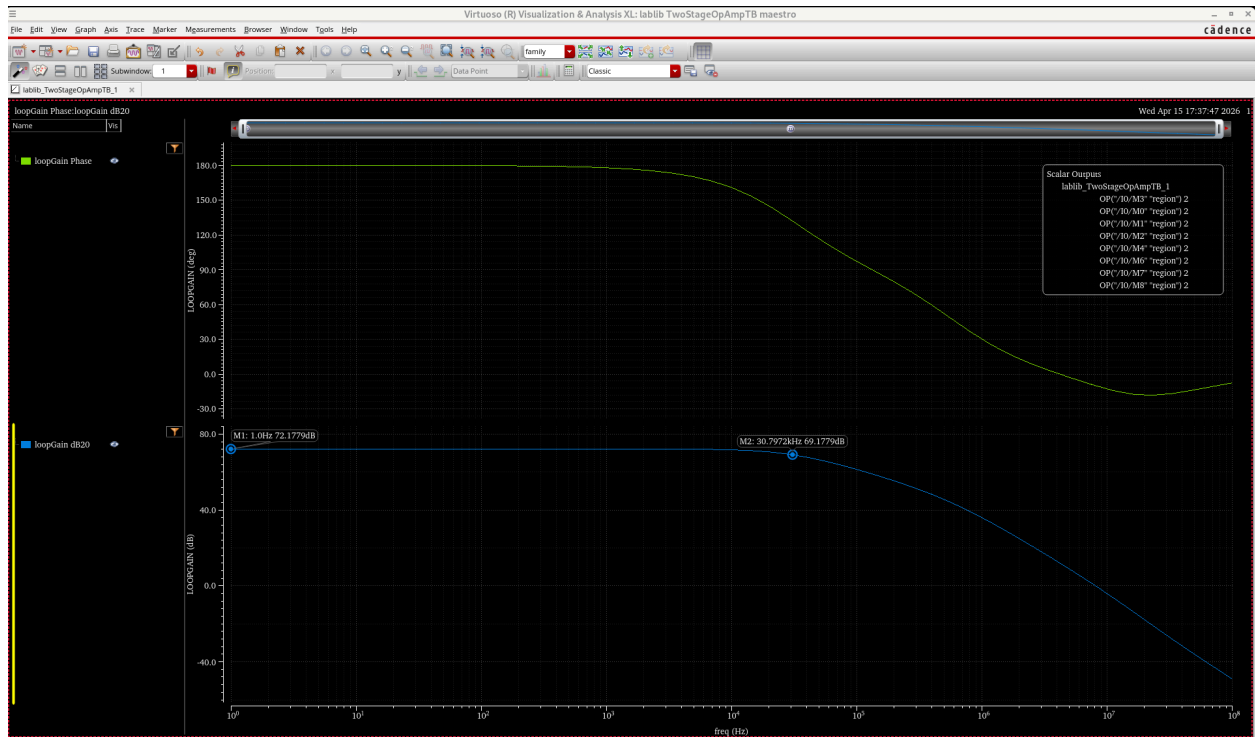
Initial 7T Op-Amp Stability Testbench

Using maestro to test, a DC analysis is performed with region operational details specified in the output setup. This confirms that each transistor is in saturation, providing assurance that the op amp is sized and wired correctly.

Name	Type	Details	Value	Plot	Plot Target	Save	Spec
OPC70/M3	region		2				
OPC70/M0	region		2				
OPC70/M1	region		2				
OPC70/M2	region		2				
OPC70/M4	region		2				
OPC70/M5	region		2				
OPC70/M6	region		2				
OPC70/M7	region		2				
OPC70/M8	region		2				
loopGain Phase	expr	phaseDegUnwrapped(Data("loopGain" result "stb"))	Hz				
loopGain dB20	expr	dbmagGetData("loopGain" result "stb"))	Hz				

Saturation and Bode Plot Testing

Next, a stability (stb) analysis is defined to sweep from ~DC to 100 MHz, providing the following open loop phase (green) and gain (blue) traces.



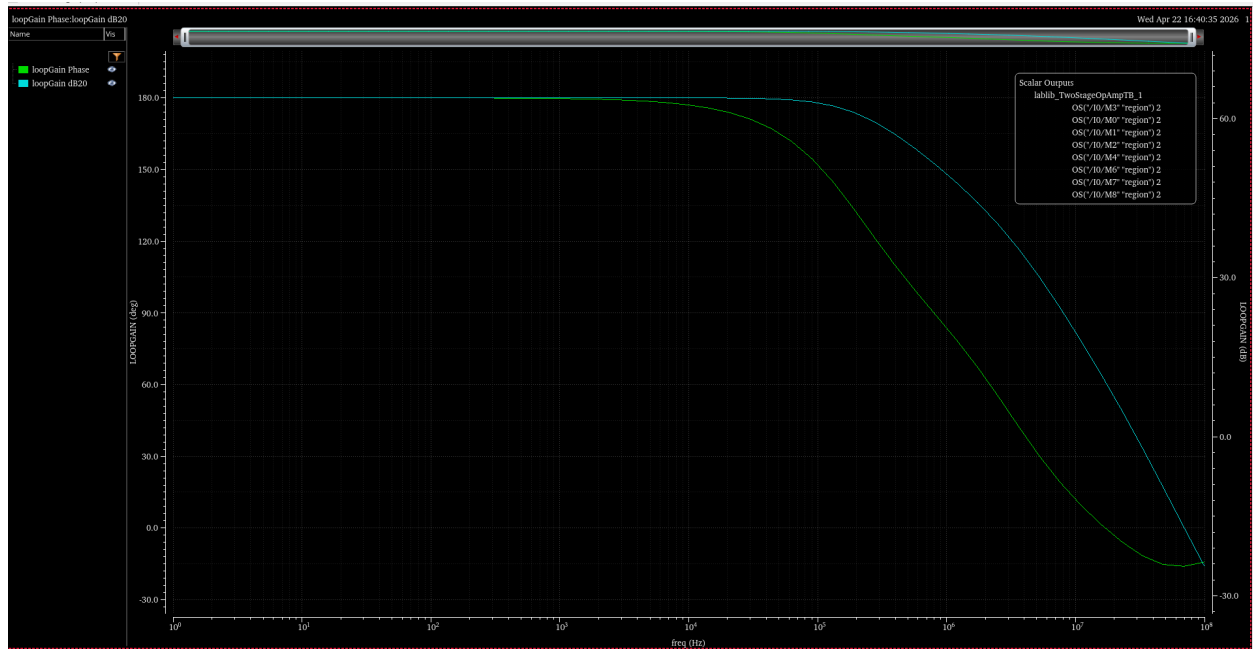
Open Loop Gain (Ideal: 3V +1kOhm)

The DC gain is 72.18 dB, and it rolls off by 3 dB at 30.8 kHz. This point represents a sort of cumulative effective or dominant pole of the op amp as a whole. Gain-band width can now be calculated.

$$GBW = A0 * f_{3dB} = 10^{72.18/20} * 30.8 * 10^3 = 125.18 \text{ MHz}$$

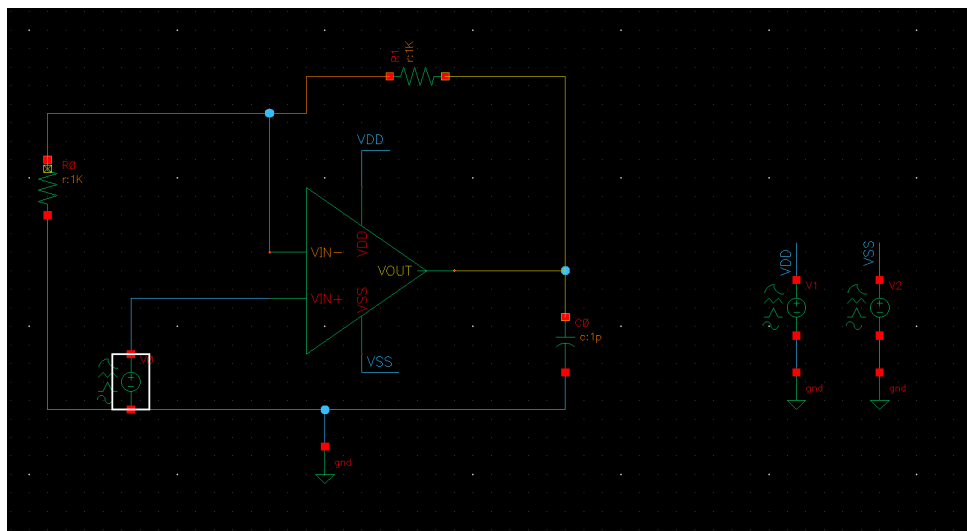
With our noninverting mode ideally reaching a gain of 16, this limits our bandwidth to about 7.8 MHz. The inverting mode will then ideally have a bandwidth of 15.6 MHz. This, along with the phase plot, is more than satisfactory for the modest requirements of this project, which was confirmed by our TA.

This is when our group realized that the voltage source in our current design was not realizable if in an eventual layout. The correction for this was changing the size of the series resistor and relying on VDD to drive the necessary current. By plotting each transistors' operating region over a swept variable resistance, it was confirmed that any resistance greater than 5.5 kOhms was adequate. The resppoly_2 resistor was chosen during this phase of our design as it was in the GPDK library and was capable of scaling to 5.5 kOhms. The open loop analysis was run again to confirm that this change didn't have detrimental effects, which it didn't.



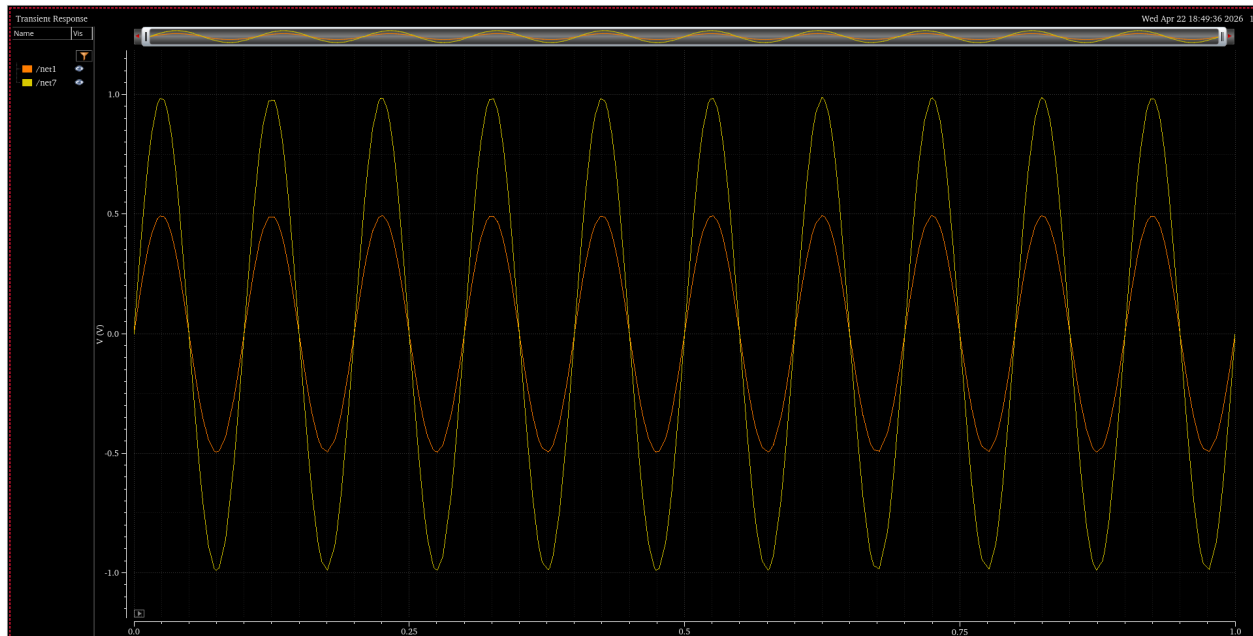
Open Loop Results (Process Resistor: resppoly_2)

The final phase of op amp verification involved testing the circuit in the inverting and non inverting configurations that would be used in our first two modes of functionality. First, mode 0 was simulated by placing the input signal on the non-inverting terminal. Using the architecture that is learned in introductory circuits courses, we set both resistors to 1 kOhms.



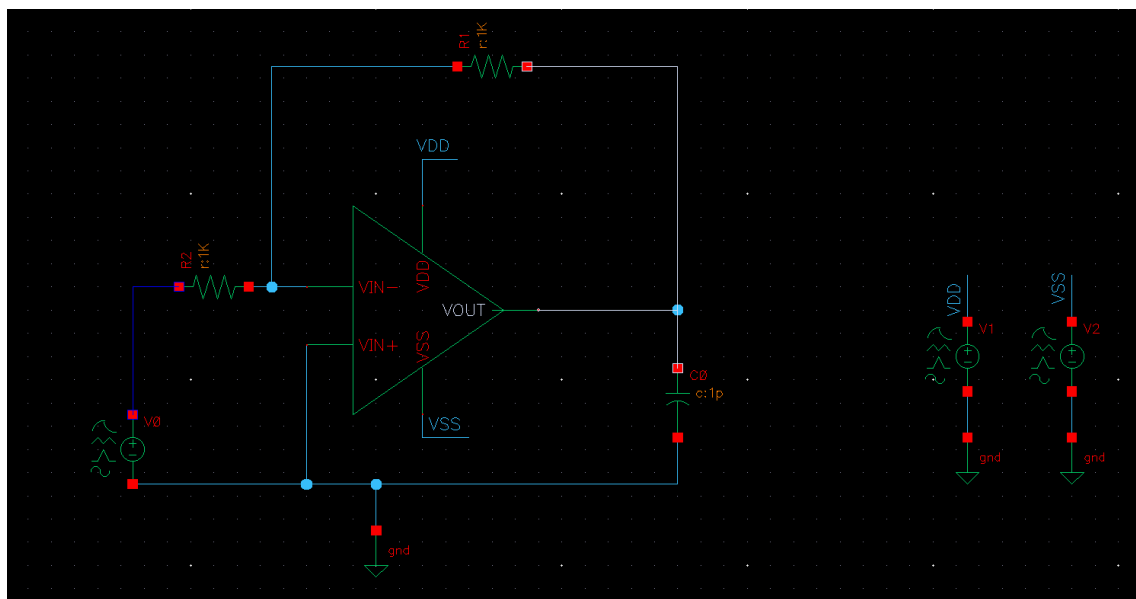
Non Inverting Op Amp Test bench Schematic

The expected gain of this setup is 2 ideally. Feeding in a 500 mV AC input, VOUT clearly reads just about a 1 V peak and is in phase with the input.



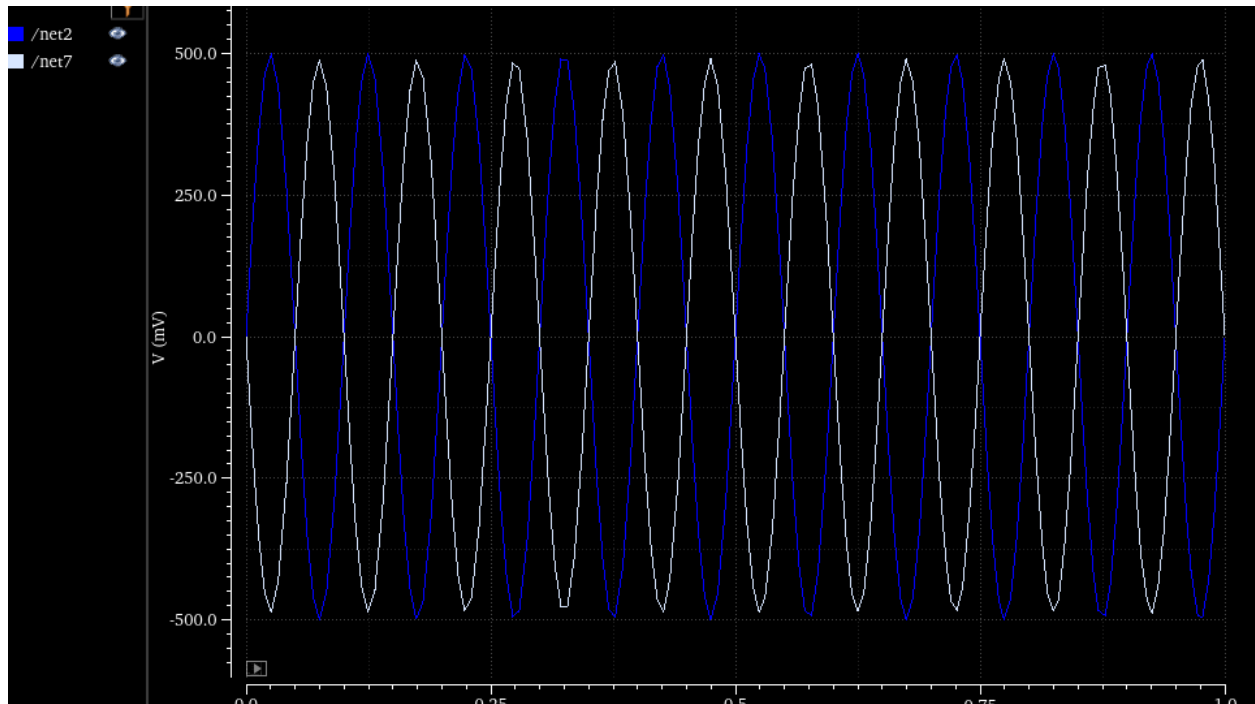
Non-inverting Test bench Results

Doing the same analysis for a fundamental inverting architecture, the mode 1 test bench was made.



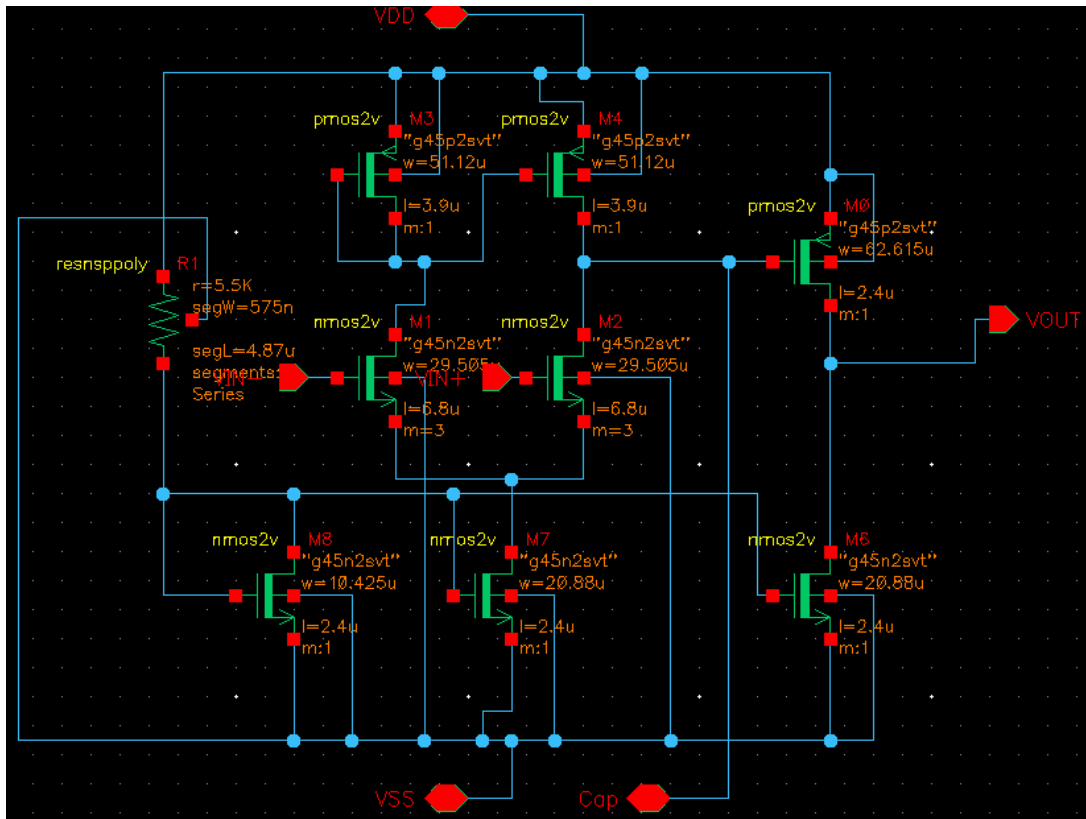
Inverting Op Amp Test Bench

The output should result in unity gain with a 180 degree phase shift, and the simulation agrees with the theory.



Changing the resistor ratios to reach the maximum gain necessary for our functionality showed no signs of clipping. This officially verifies that the op-amp will operate under the conditions that we plan to put it through in our project.

We decided at this point to swap out the internal compensating capacitor for an external one, simplifying layout and adding more degrees of freedom for the use. Adding an input-output pin for this connection, the finalized schematic before layout was set.

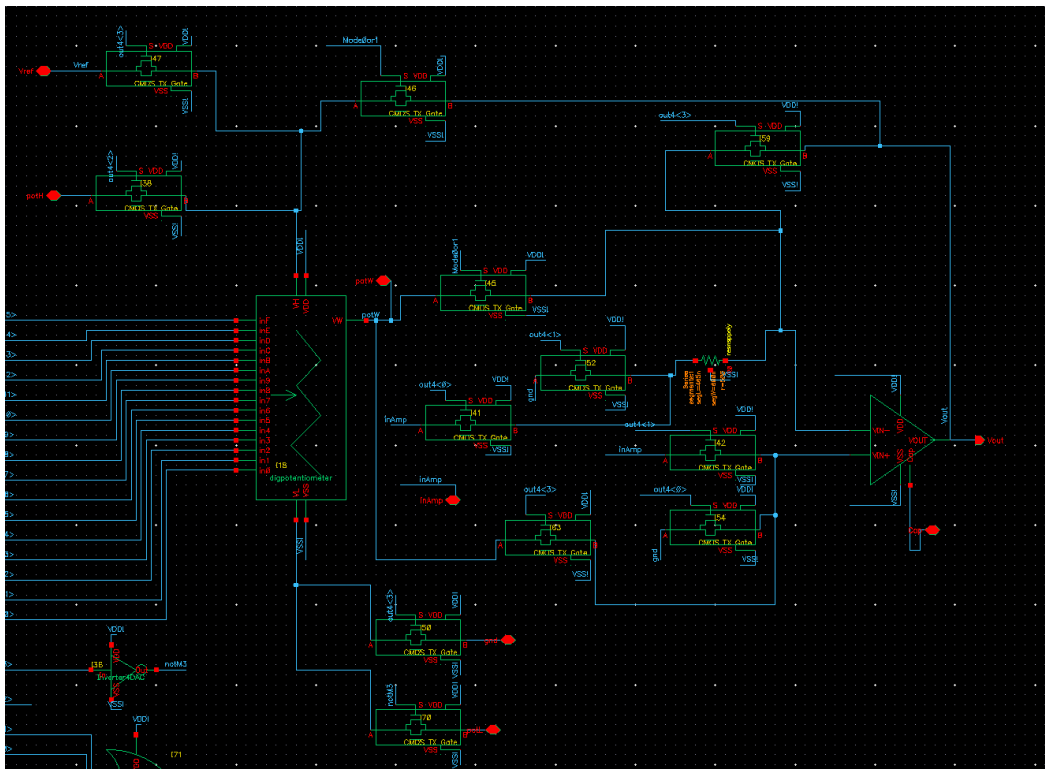


Finalized Op Amp Schematic with External Capacitor

3. Final Integrations and Functional Verification

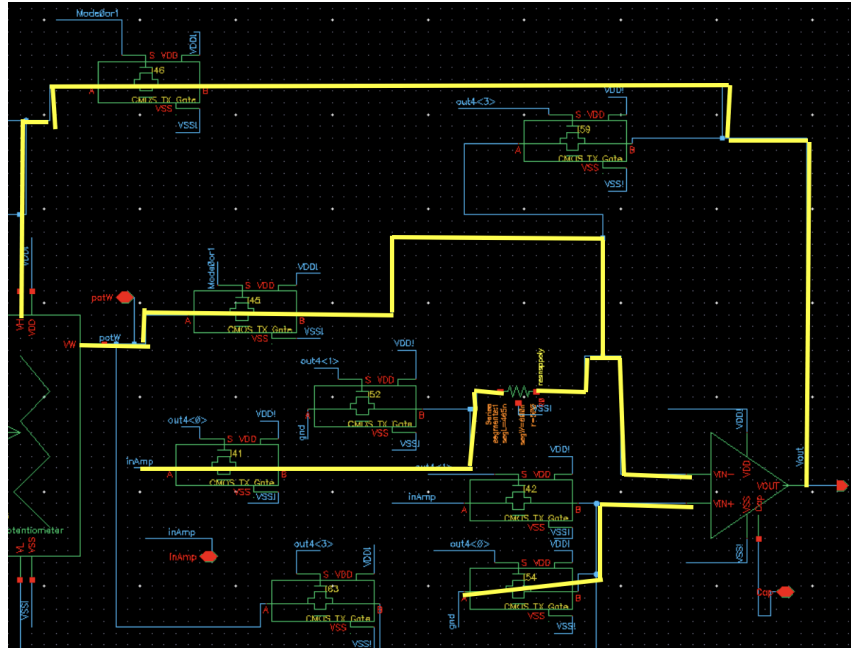
With all fundamental schematics and integrations tested, constructed and tested, our next step was to proceed with our full circuit integration.

To recap our design plans, including our new additions, we ran through our mental map of what the low-level behavior of our circuit does. The digital potentiometer serves as the central scaling element within the system. It is used in every single operation of this device, ranging from gain control to variable resistance control. The transmission gate matrix facilitates the routing of specific voltages from these taps based on the state of the 4-16 decoder. On the right side, CMOS transmission gates can be seen reconfiguring the op-amp circuit for its use in three of the four modes.



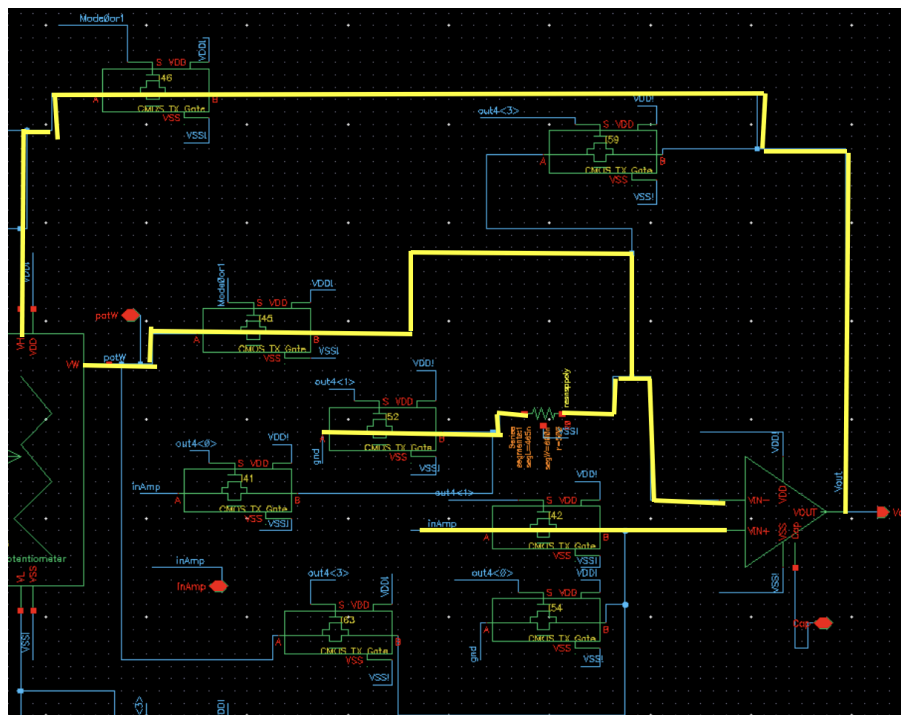
Op Amp Circuit Reconfiguration System

To allow three different operations of the op amp, we needed to create a system which electrically reroutes the op amp feedback and input networks, using outputs from our decoder module. This meant using CMOS transmission gates to switch different paths on/off to achieve our functions. These paths can be visualized in the following graphics.



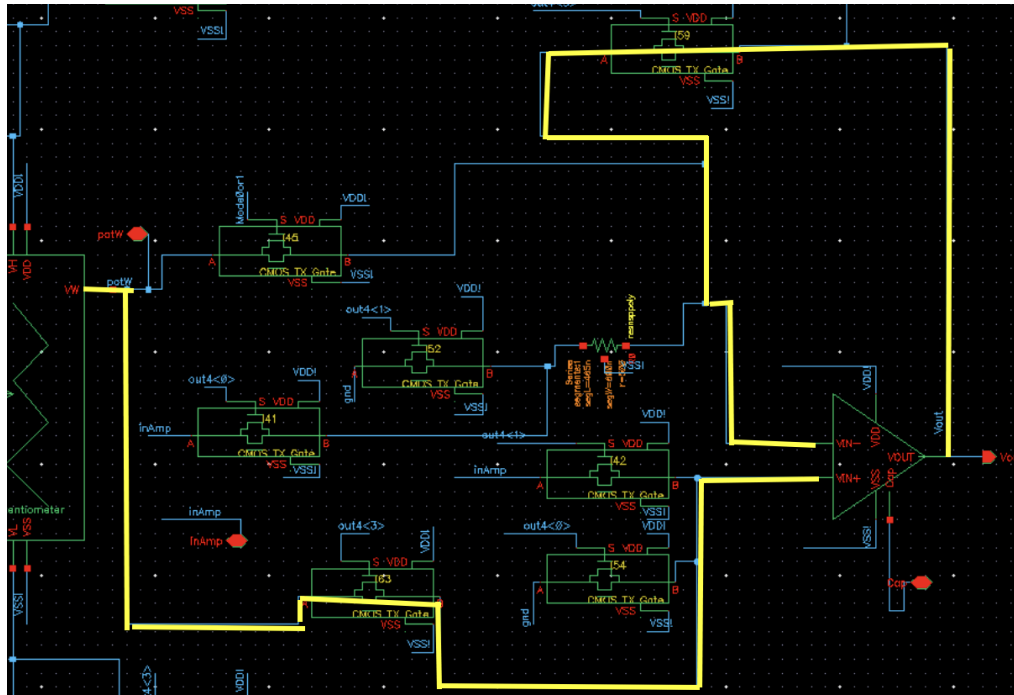
Mode 0 Path

Mode 0 features a negative feedback network containing the potentiometer wiper resistance, with a grounded non-inverting terminal and the output being fed through a 500-ohm resistor into the inverting terminal. This causes inverting amplification.



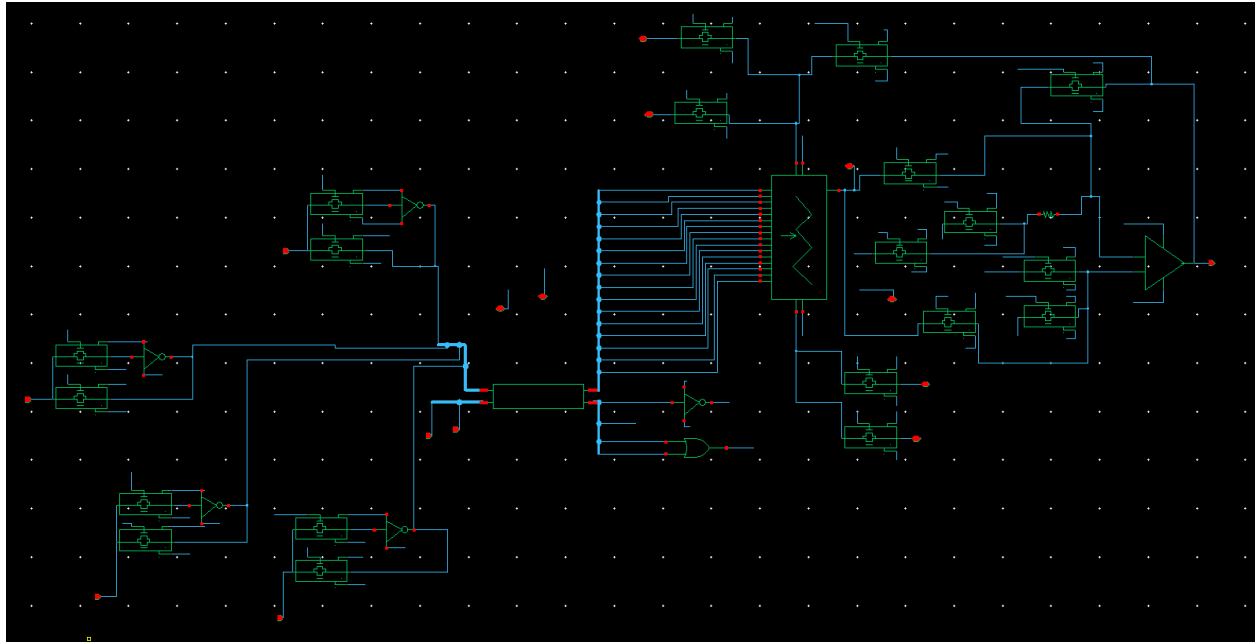
Mode 1 Path

Mode 1 features a negative feedback network containing the potentiometer's wiper resistance, with the inverting terminal grounded through a 500-ohm resistor and the input signal being connected to the non-inverting terminal. This causes non-inverting amplification.



Mode 3 Path

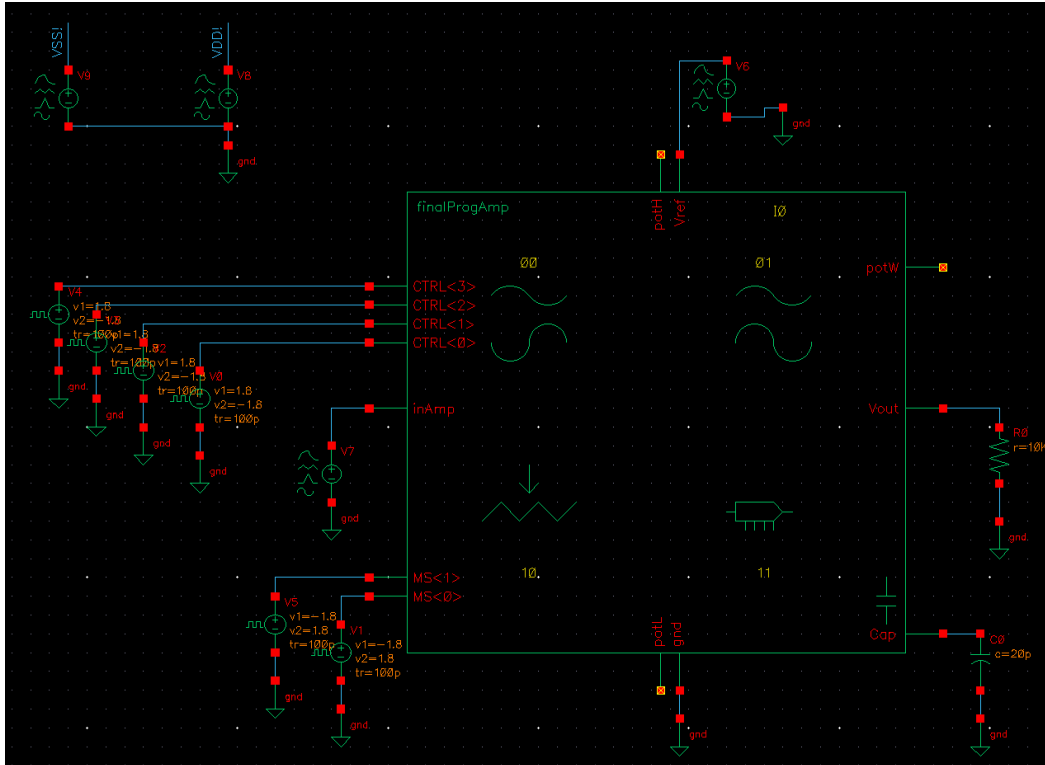
In the mode 3 case (DAC), the wiper voltage output, which is the output of the voltage divider configured by the potentiometer and therefore the desired DAC output level, is connected to the non-inverting terminal of the op amp. The negative feedback network in this mode is configured as a short, meaning that the op amp is acting like a unity gain buffer, buffering the DAC output.



Programmable Amp/DigiPot/DAC Schematic View

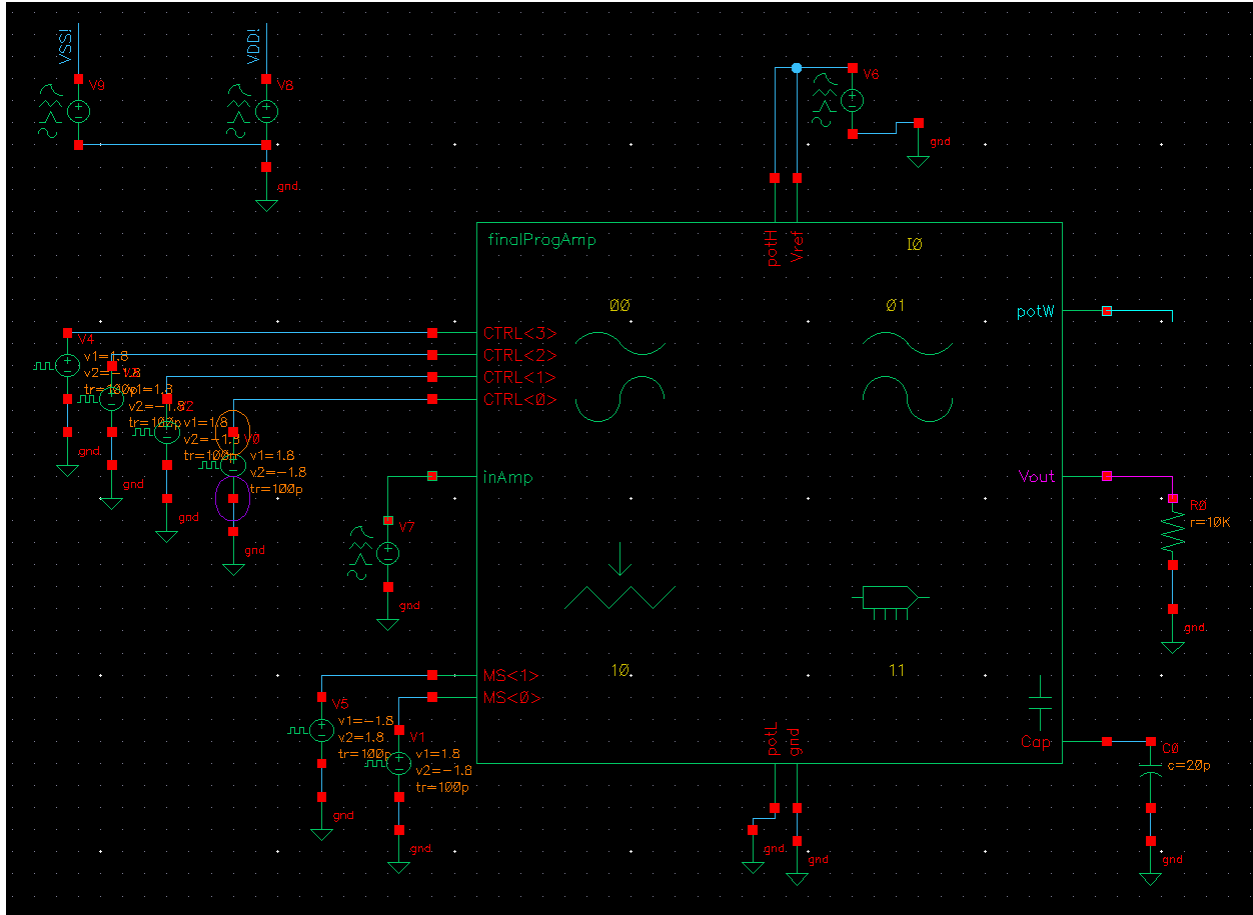
With our mental model for the circuit's structure explored, we tested the full system in all 4 modes to confirm that this model matched the circuit that we have virtually assembled. We created a symbol for the device that clearly indicates four modes of operations. Setting both the 4-bit wiper control and the mode control to cascading digital switching inputs like we have done prior, we can test every wiper position for each mode in one simulation.

Creating a testbench for the top-level module was fairly straightforward: using voltage sources from analoglib to cycle through the modes of operation at intervals of 8ms, with each of the 4-bit gain control inputs increasing by one LSB every 0.5ms, allowing the full range of inputs to be tested for each mode. From there, a resistive load was added (arbitrarily selected to be 10k ohms), the voltage supplies for VDD (1.8V) and VSS (-1.8V) were included from analoglib, and the reference voltage (1V) was connected to the V_ref port for DAC operation.



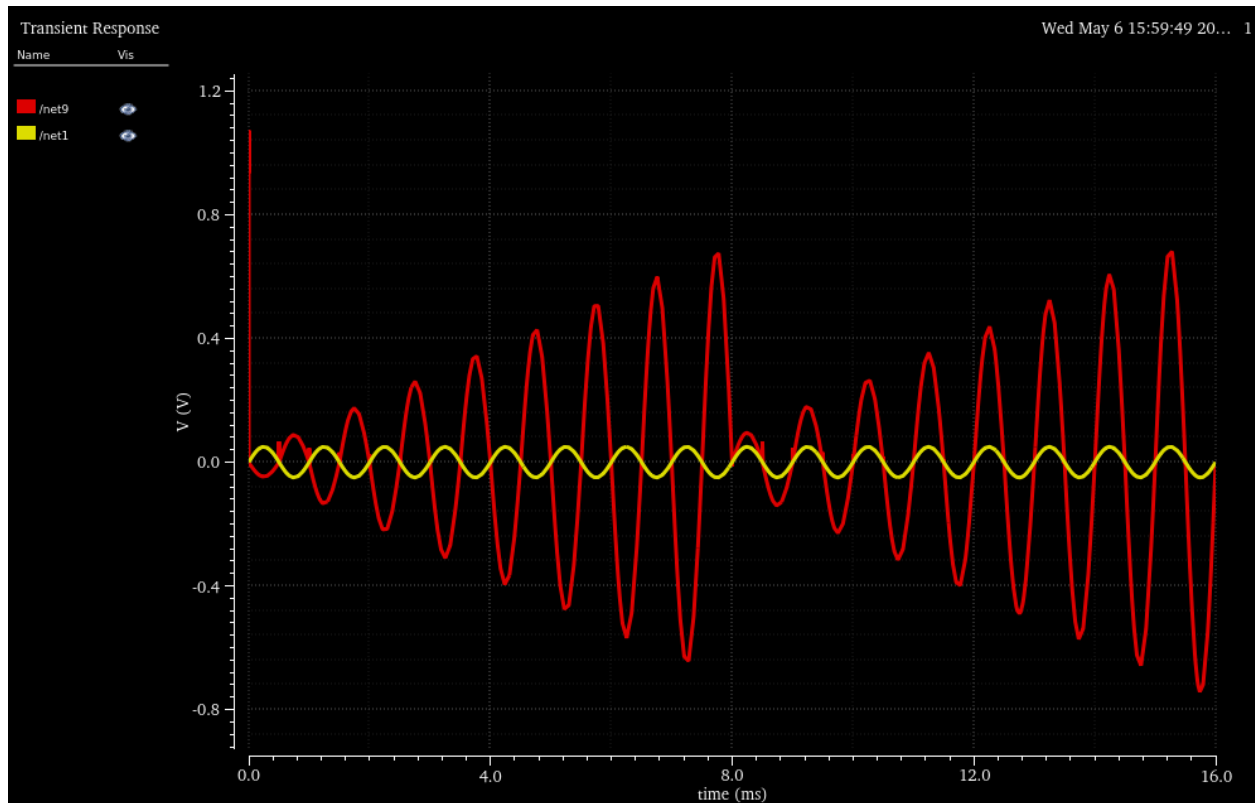
Test Bench for Programmable Amp/DigiPot/DAC

This testbench configuration was useful for testing three of the four modes, as the potentiometer mode (mode 2) would require a slightly different configuration for testing. To test the potentiometer mode, we could use two different approaches: measuring current that passes through the wiper terminal when driving a load at that terminal, or measuring the wiper voltage while grounding the V_L terminal. The first configuration allows us to measure the potentiometer function by observing the change in resistance between the high and wiper terminals of the potentiometer, which will increase (thus current decreasing) as the 4-bit control input increases. The other testing configuration will show a decrease in the wiper voltage as the 4-bit control input increases, as the wiper will move down the resistor ladder, causing a smaller resistance between the wiper and ground.



Test Bench for Mode 2 (Potentiometer)

The testbench we created successfully produced the results we sought, showing that our circuit schematic was functionally correct. Due to the testing difficulty of Mode 2, the testbench had to be run thrice: once in the original configuration for 16ms (modes 0 and 1), once in the original configuration with the mode select adjusted for 8ms (mode 3), and once in the modified configuration for 8ms (mode 2). In total, the simulations provided 32ms of simulated data, which cycled through all possible input configurations.



Mode 0 & 1 Test Results

The results shown above demonstrate the amplifying capabilities of this device, with both inverting amplification (mode 0) and non-inverting amplification (mode 1). For each half cycle of the sinusoidal input/output, the gain is increasing by one step.

In the case of Mode 0, it can be seen that the input signal (yellow) and output signal (red) are out of phase by 180 degrees, exhibiting an inverting gain, which is dependent on a 500-ohm series resistor with the input and the potentiometer wiper resistance in the negative feedback network. Since the gain is equal to R_f/R_{in} and the wiper resistance is the feedback component, the gain increases as the 4-bit gain control input increases.

Step	Output Mag Pk (mV)	Gain
1	47	0.94
2	89	1.78
3	135	2.7
4	175	3.5
5	218	4.36
6	261	5.22
7	311	6.22
8	343	6.86
9	395	7.9
10	427	8.54
11	475	9.5
12	505	10.1
13	568	11.36
14	599	11.98
15	643	12.86
16	675	13.5

Mode 0 Gain Measurements

As listed in the table above, the 16 different levels of gain that we desired are present, exhibiting mostly linear behavior as the magnitude of the gain increases from 0.94 to 13.5. Ideally, this gain magnitude range would be 1 to 16, if the potentiometer and op-amp were ideal, but due to non-idealities, such as the CMOS transmission gate series resistance and other non-idealities associated with the op-amp, the gain levels are slightly lower than those ideal values. Either way, this device provides 16 distinctly different levels of gain that can be chosen via the 4-bit control inputs.

In the case of Mode 1, it can be seen that both the input signal and output signal are in-phase, exhibiting non-inverting amplification behavior. In this case, the 500-ohm resistor is connected to the inverting terminal and ground, with the feedback resistor still being the wiper resistance of the potentiometer. With the input signal now being connected to the non-inverting terminal, the gain is equal to $1 + R_f/R_{in}$, or the gain of the inverting amplifier plus one.

Step	Output Mag Peak (mV)	Gain
1	95	1.9
2	140	2.8
3	172	3.44
4	229	4.58
5	264	5.28
6	316	6.32
7	354	7.08
8	398	7.96
9	438	8.76
10	489	9.78
11	524	10.48
12	572	11.44
13	607	12.14
14	657	13.14
15	681	13.62
16	743	14.86

Mode 1 Gain Measurements

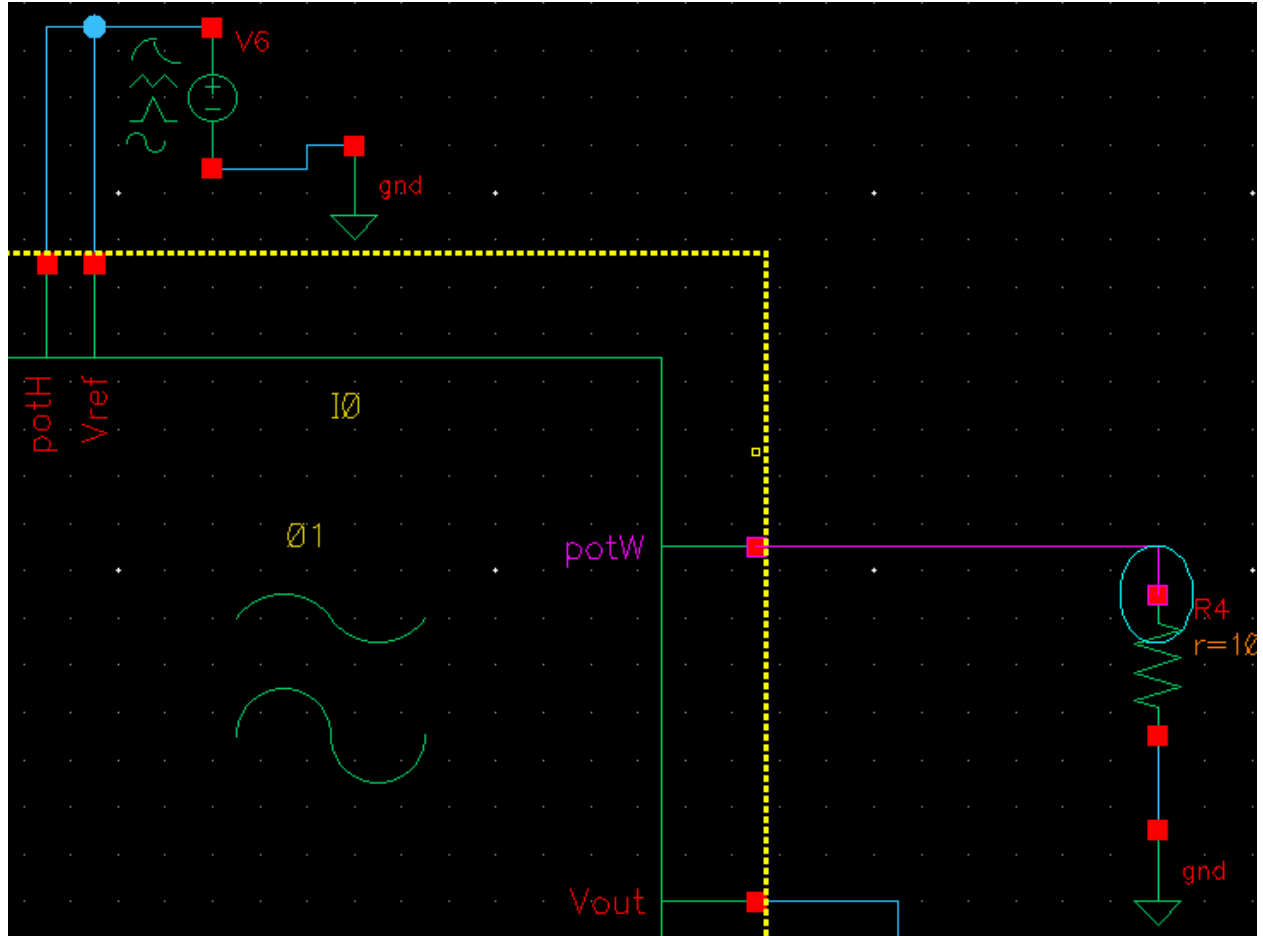
As seen in the table above, gain increases mostly linearly as the 4-bit input increases, reaching 16 distinct levels of gain, ranging from 1.9 to 14.86. It can be seen that these levels of gain are all roughly 1 V/V higher than those of Mode 0, which is to be expected in the non-inverting configuration containing the same resistors, per the formula mentioned previously.

Next, to test the operation of Mode 2, using the potentiometer as a standalone potentiometer, we used the two styles of testing mentioned previously in this report: measuring current and measuring wiper voltage in a voltage divider configuration.

First, we tested the potentiometer by grounding the V_L terminal, connecting 1V to V_H , and measuring the wiper voltage. As expected, the wiper voltage decreased by evenly-spaced increments, ranging from 935.7 mV to 1.5 mV as the 4-bit input increased. Ideally, this would range from 937.5 mV to 0 V, so our operation was incredibly close to the ideal behavior. The results from this test can be found below.

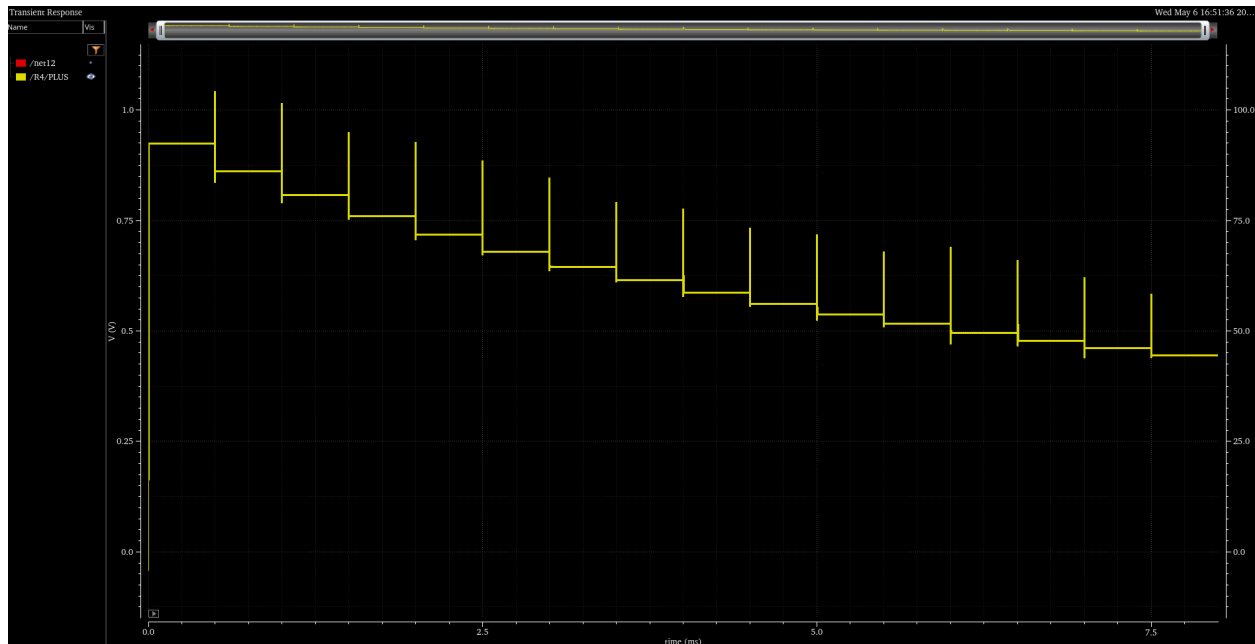


Mode 2 Results (Grounded V_L ; $V_H = V_{ref}$)



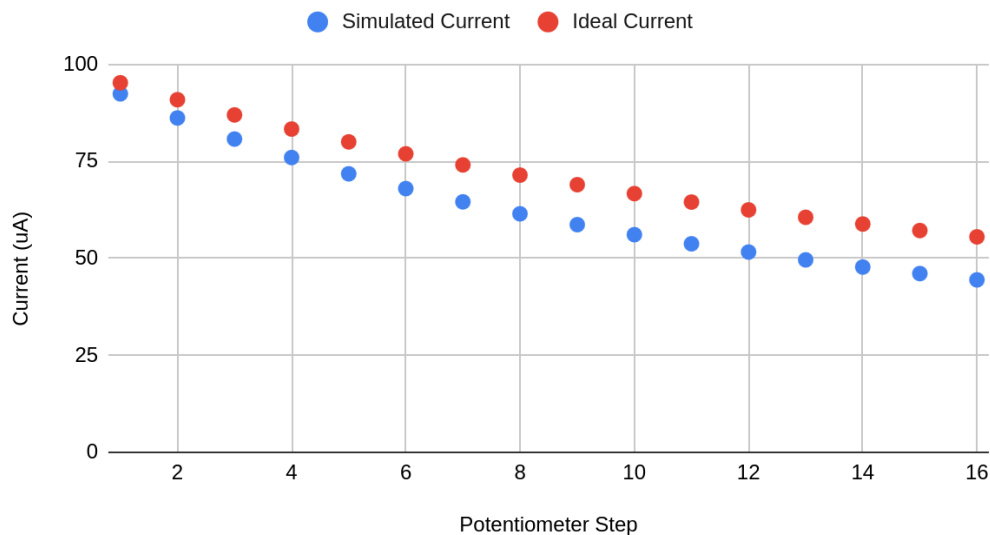
Mode 2 Current Testbench

The next method of testing Mode 2 was measuring current (as seen above) between the V_H and wiper terminals, leaving V_L floating, effectively turning the potentiometer into a variable resistor. For this test, we added a 10k ohm resistor to the wiper terminal, which caused the shift in current as the 4-bit input increases to be slightly non-linear. This is because the series resistance in this testbench is equal to $10k + R_{Wiper}$. The results can be seen below, along with a plot comparing this non-linear behavior to the ideal result of $\frac{1}{10k + R_{Wiper}}$. Overall, there is high accuracy between the two.



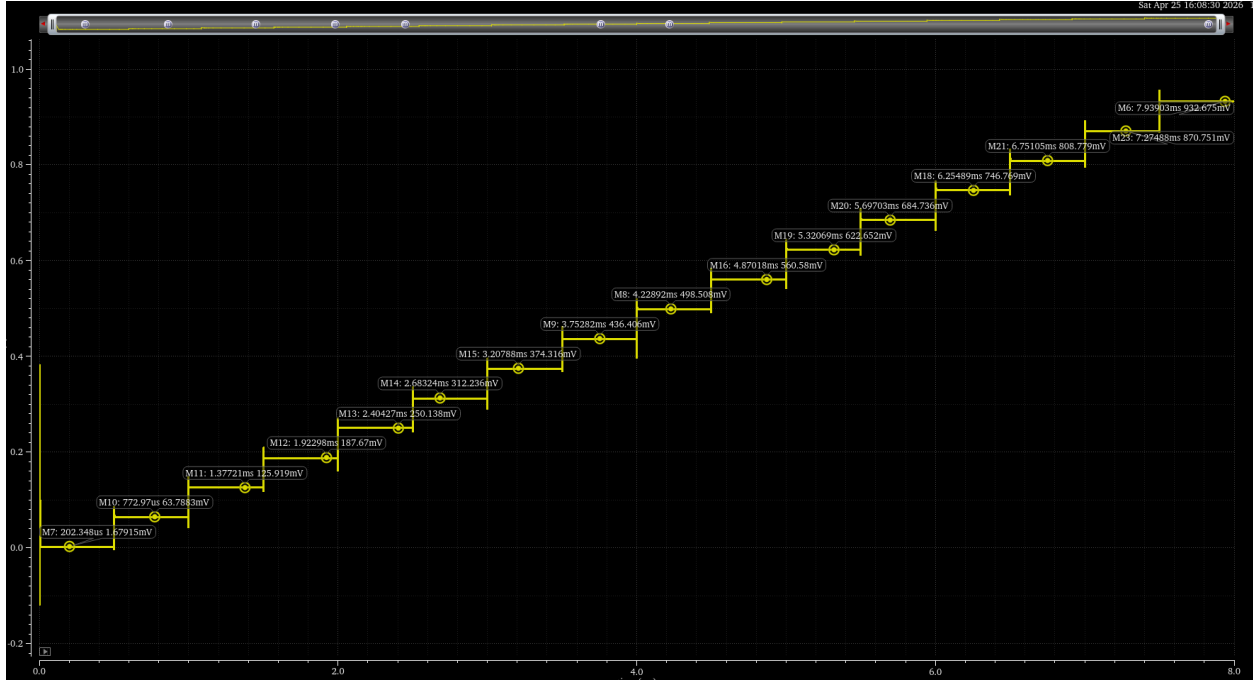
Mode 2 Results (Floating V_L; V_W connected to 10k load)

Current vs Potentiometer Step (w/ 10k load)



Mode 2 Results Compared to Ideal Current Behavior

The final mode to assess was Mode 3: the buffered DAC. The output of the DAC ideally increases by one LSB (in this case, 62.5 mV) with every increase of the 4-bit input, meaning that these results should ideally very closely resemble the results from the voltage-divider test bench of Mode 2, but with the direction flipped: increasing rather than decreasing over time.



Mode 3 Test Results

Step	Output (mV)	Output Level	DNL
1	1.67915	0.17%	
2	63.7883	6.38%	-0.0062536
3	125.919	12.59%	-0.0059088
4	187.67	18.77%	-0.011984
5	250.138	25.01%	-0.000512
6	312.236	31.22%	-0.006432
7	374.316	37.43%	-0.00672
8	436.406	43.64%	-0.00656
9	498.508	49.85%	-0.006368
10	560.58	56.06%	-0.006848
11	622.652	62.27%	-0.006848
12	684.736	68.47%	-0.00656
13	746.769	74.68%	-0.007472
14	808.779	80.88%	-0.00784
15	870.751	87.08%	-0.008448
16	932.675	93.27%	-0.009216

Mode 3 Test Output and Differential Non-linearity

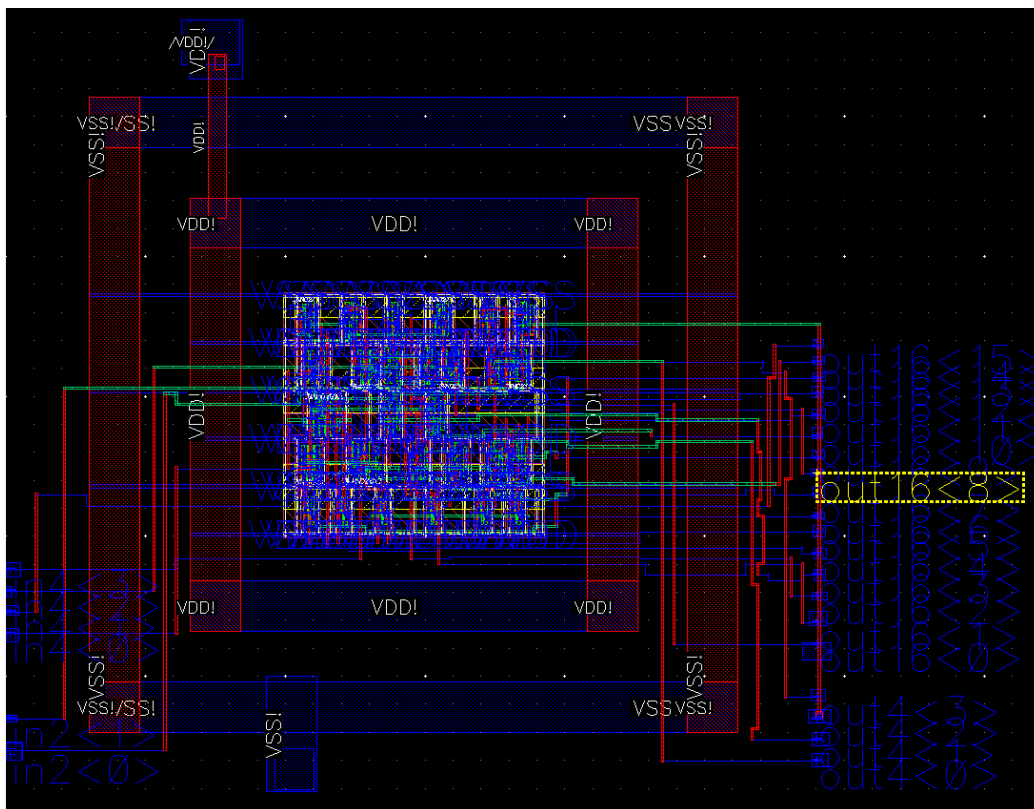
It can be seen in the above figures that the behavior of our DAC mode was highly linear, with a range of 1.68 mV to 932.68 mV, and a worst-case differential non-linearity of just 1.2% inaccuracy between Steps 3 and 4. Differential non-linearity is a metric used to assess DACs, by calculating the deviation between the two analog levels of two adjacent digital values compared to the ideal difference. In this case, it compares the step size to 62.5 mV, which is an ideal step (1/16 of V_{ref}). Other than the worst case, the DNL remains under 1% for every other step, maintaining low error.

Overall, each of our modes showed very satisfactory results, effectively integrating each of the required functions into our device. Each of these functions is slightly non-ideal, but regardless they all work correctly, which we are proud of.

4. Layout and Physical Verification

With our schematics verified, our focus shifted towards translating our ideal parameters into a realizable layout. As is standard in our class and in industry, to ensure that our cell placement is both correctly wired and sized in a manufacturable way, all of our components are run through DRC and LVS.

Starting with our multi-functional decoder, the layout was rather simple due to it being written in verilog initially, allowing automatic layout via Cadence Innovus. With just a few keystrokes and some menu selections, we were able to have every optimal logic gate and therefore transistor already laid out and routed. The only work required in Layout XL was to add the power pins and each of the I/O pins. From there, it was immediately ready to add into our top-level layout.



Decoder Layout

In Innovus, before importing to Cadence Virtuoso, we were able to verify this module both geometrically and electrically. With just a few commands, Innovus was able to confirm geometrical viability and connectivity correctness. Those reports are found below.

```
VERIFY GEOMETRY ..... Cells           : 0 Viols.
VERIFY GEOMETRY ..... SameNet          : 0 Viols.
VERIFY GEOMETRY ..... Wiring           : 0 Viols.
VERIFY GEOMETRY ..... Antenna           : 0 Viols.
VG: elapsed time: 0.00
Begin Summary ...
Cells           : 0
SameNet         : 0
Wiring          : 0
Antenna         : 0
Short           : 0
Overlap         : 0
End Summary

Verification Complete : 0 Viols.  0 Wrngs.

*****End: VERIFY GEOMETRY*****
*** verify geometry (CPU: 0:00:00.1 MEM: 164.4M)
```

Geometric Verification for Decoder

```
e make sure it is assigned and rerun verifyConnectivity.
**WARN: (IMPVFC-97):   IO pin in4[2] of net in4[2] has not been assigned. Please
e make sure it is assigned and rerun verifyConnectivity.
**WARN: (EMS-27):     Message (IMPVFC-97) has exceeded the current message display
limit of 20.
To increase the message display limit, refer to the product command reference manual.

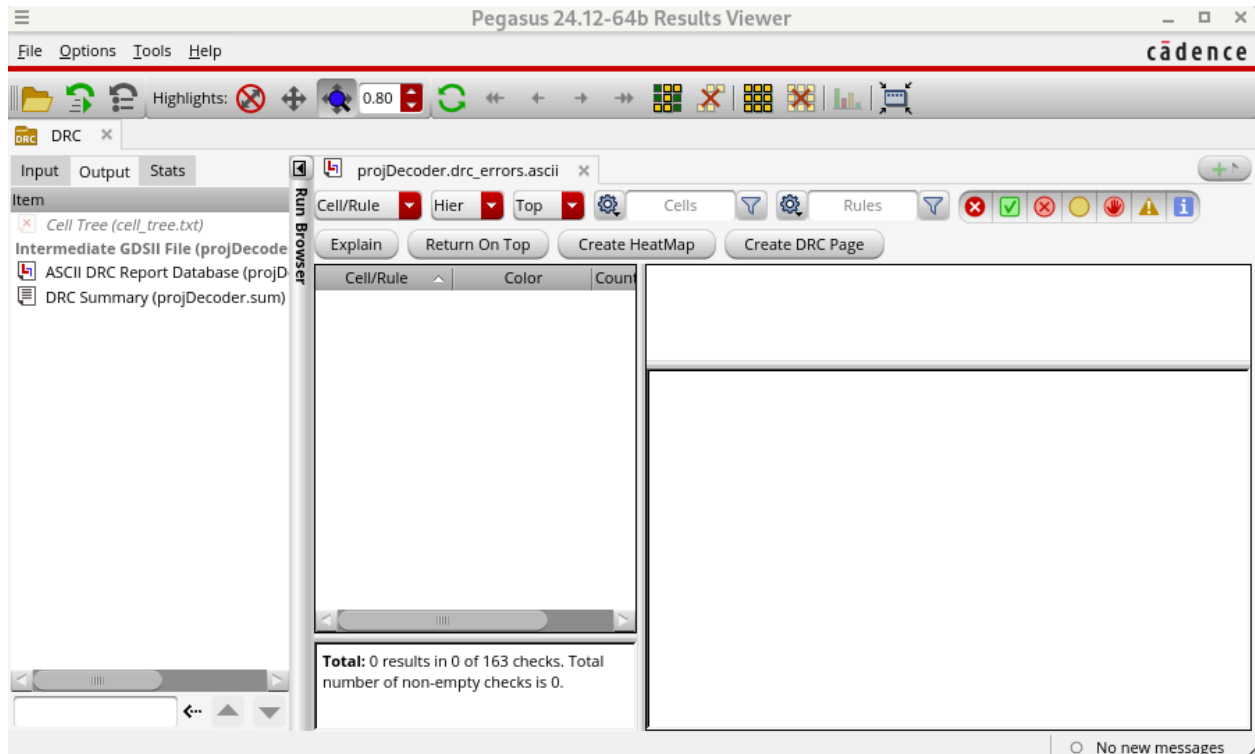
Begin Summary
Found no problems or warnings.
End Summary

End Time: Sat Apr 25 15:12:50 2026
Time Elapsed: 0:00:00.0

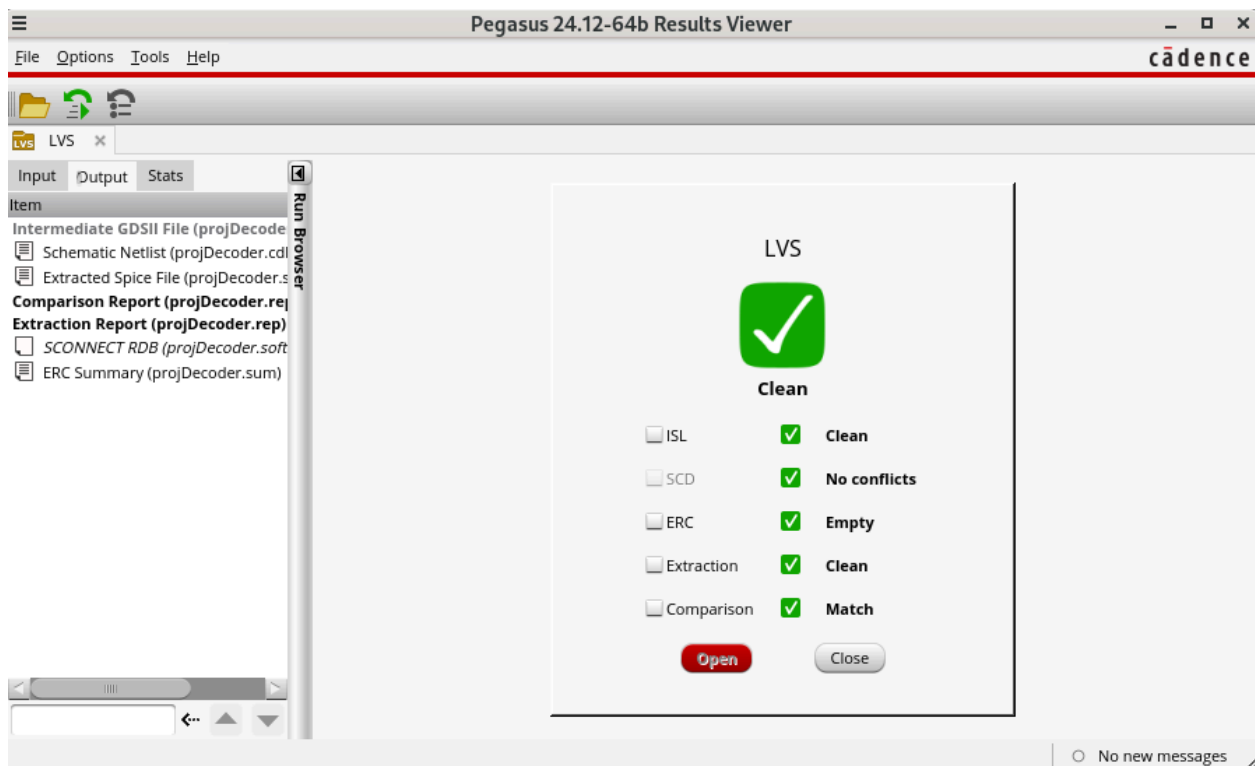
***** End: VERIFY CONNECTIVITY *****
Verification Complete : 0 Viols.  0 Wrngs.
(CPU Time: 0:00:00.0 MEM: 0.156M)
```

Connectivity Verification for Decoder

Our successful design rule and net checks are confirmed below, run in Pegasus DRC and LVS, indicating that there are no issues with our layout.

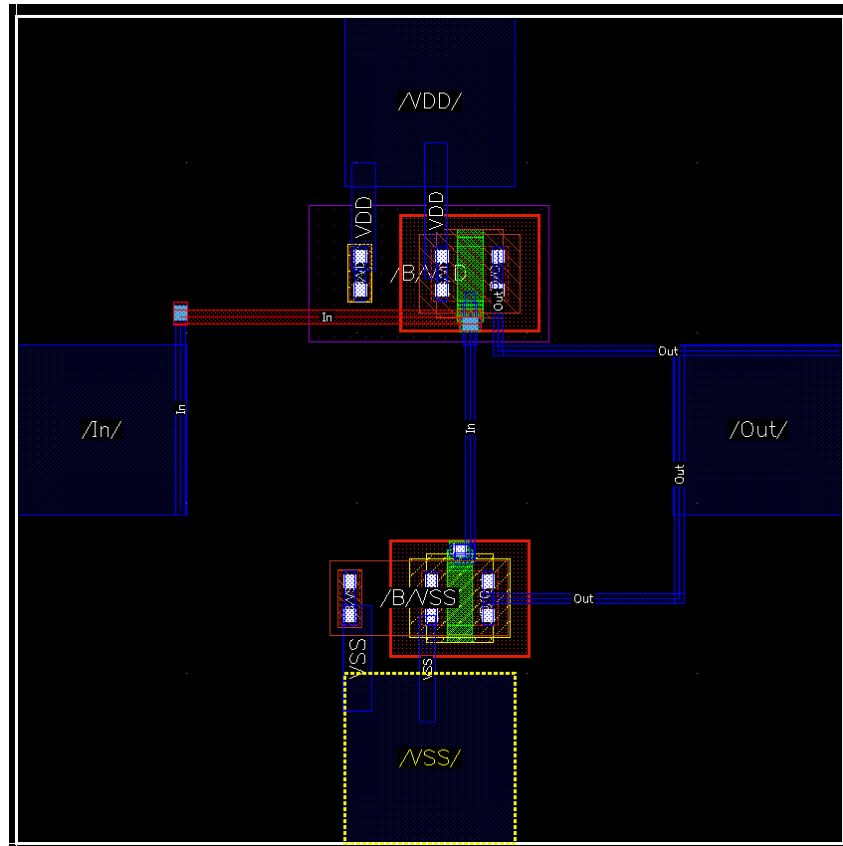


Decoder DRC Pass



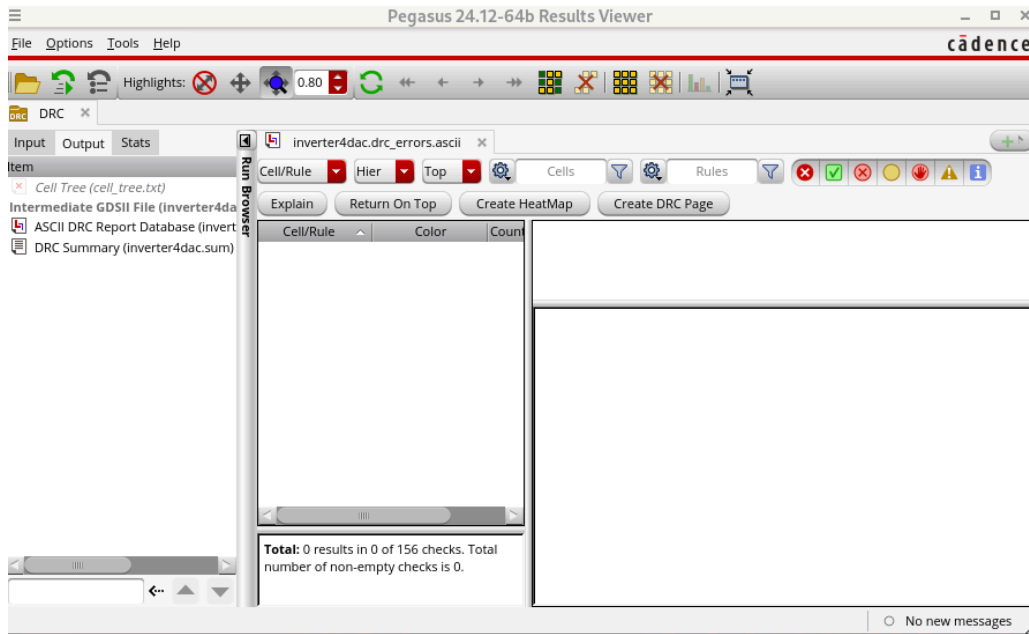
Decoder LVS Pass

Next in layout was our inverter. An inverter is found in the standard GPDK045 library, but implementing that cell into our larger circuit created some difficult-to-debug connectivity errors, so we made our own inverter, which was a very simple layout.

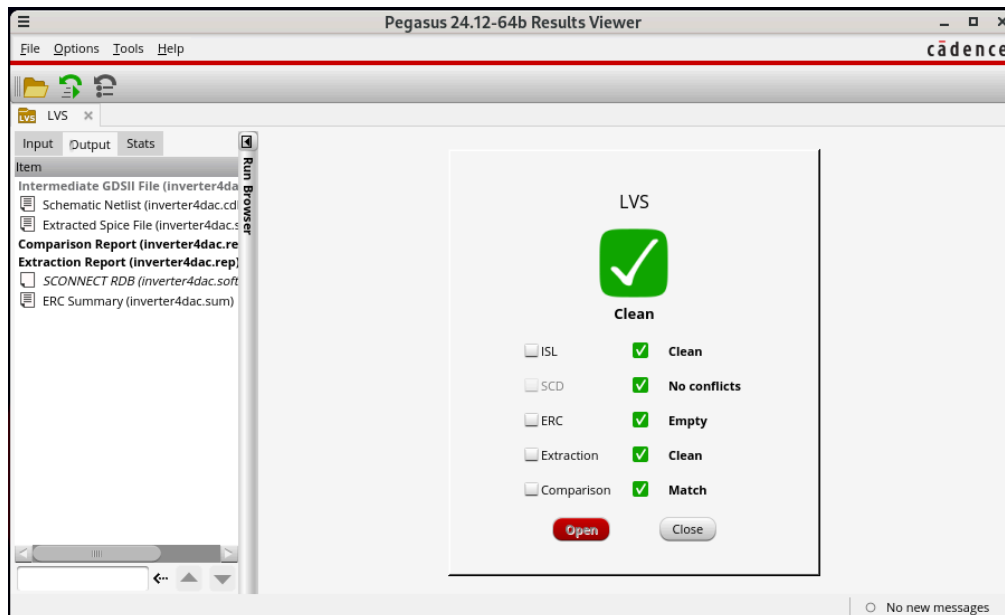


Inverter Layout

This cell similarly passed DRC and LVS with no issues.

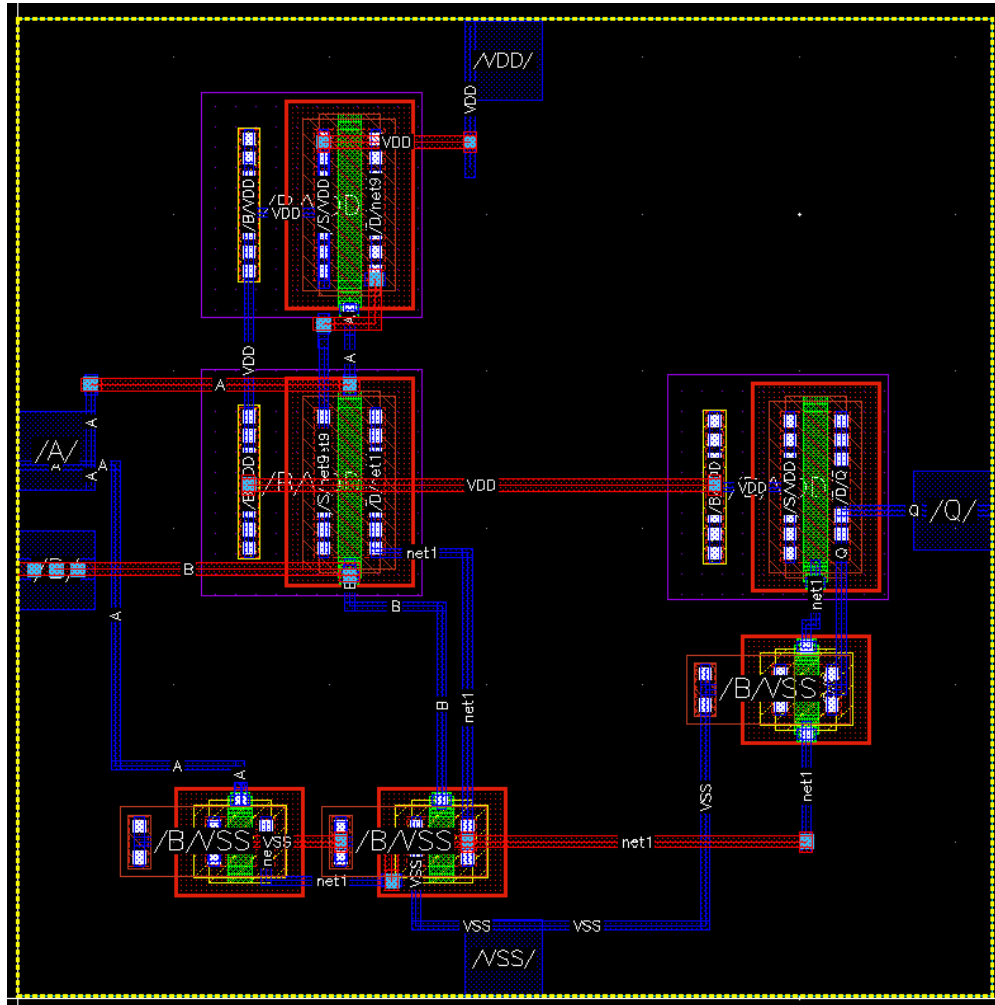


Inverter DRC Pass

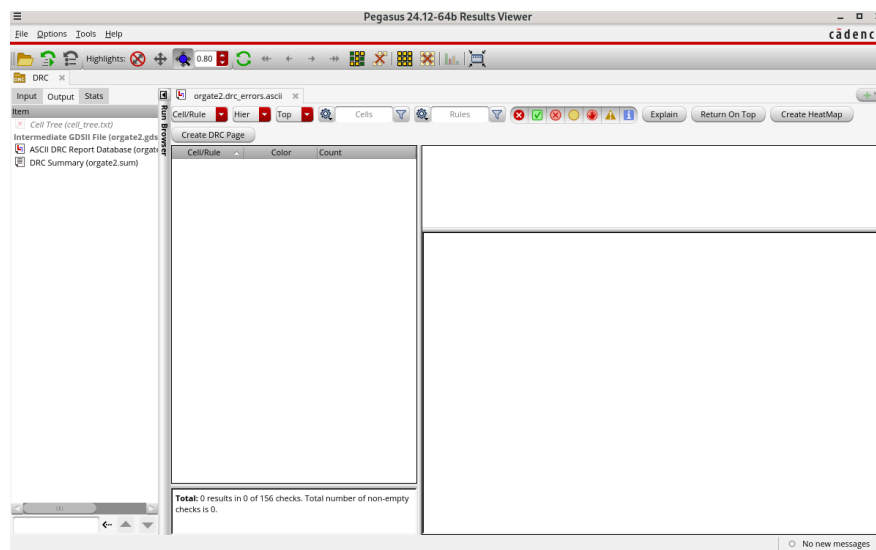


Inverter LVS Pass

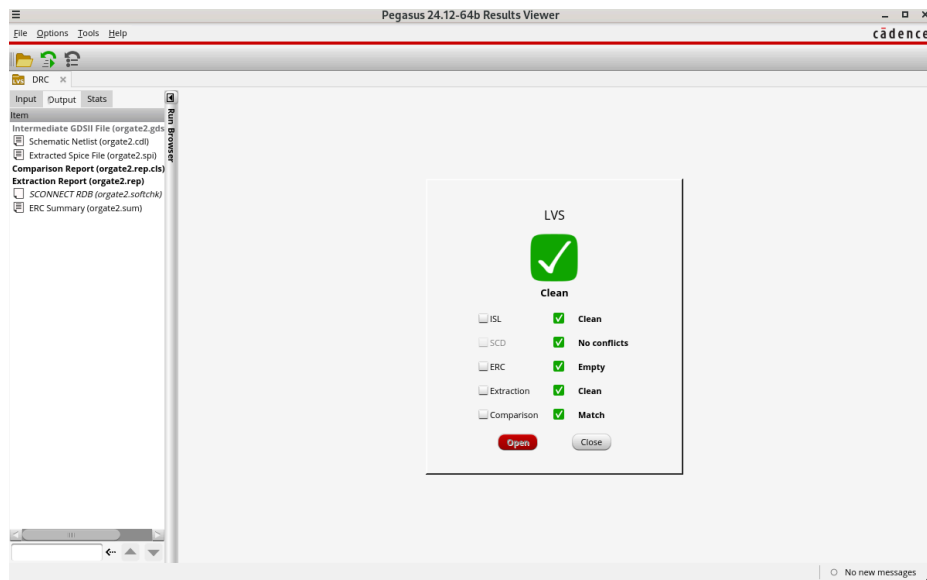
The other very simple logic gate required for our device was a basic two-input OR gate. Again, one of these could be found in the cell library, but it caused issues in layout, so we created our own and the layout process was very simple for that, as it was only six transistors. This layout can be seen below, and it passed both DRC and LVS without any issues.



Two Input OR Gate Layout

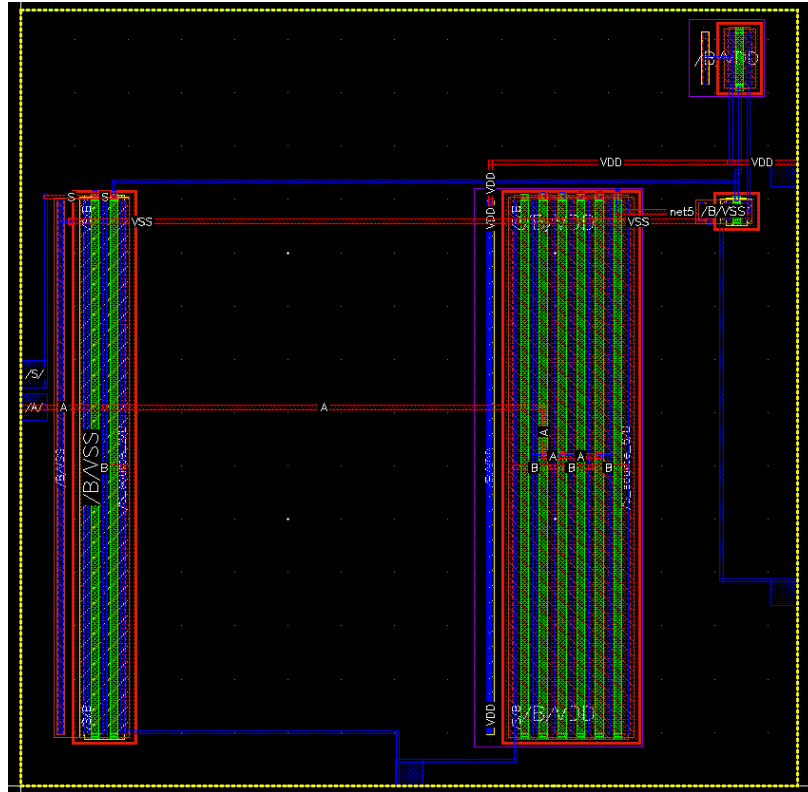


2 Input OR Gate DRC Pass



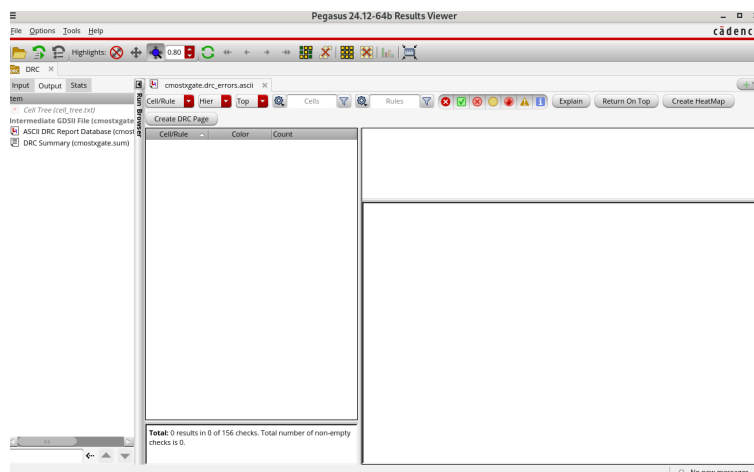
2 Input OR Gate LVS Pass

The transmission gate was also a relatively easy layout. The large driving transistor pair clearly dominates the layout, with the NMOS effective width being 20 μm and the PMOS 60 μm . Implementing fingers on these transistors cut them to a manageable layout width of only about 10 μm , while still maintaining the effectively very large width. This requires a little more intricate handling of the routing, but auto routing this cell was fairly straightforward, aside from the common issues (such as minimum spacing between vias and metal traces being violated), which are simple to fix.

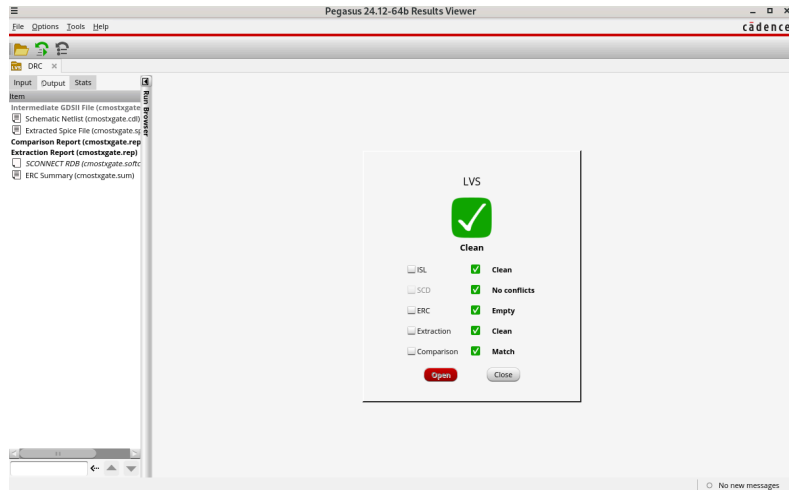


CMOS Transmission Gate Layout

As with before, we allow DRC and LVS checks to systematically confirm our component layout. Both of these passed without any issues

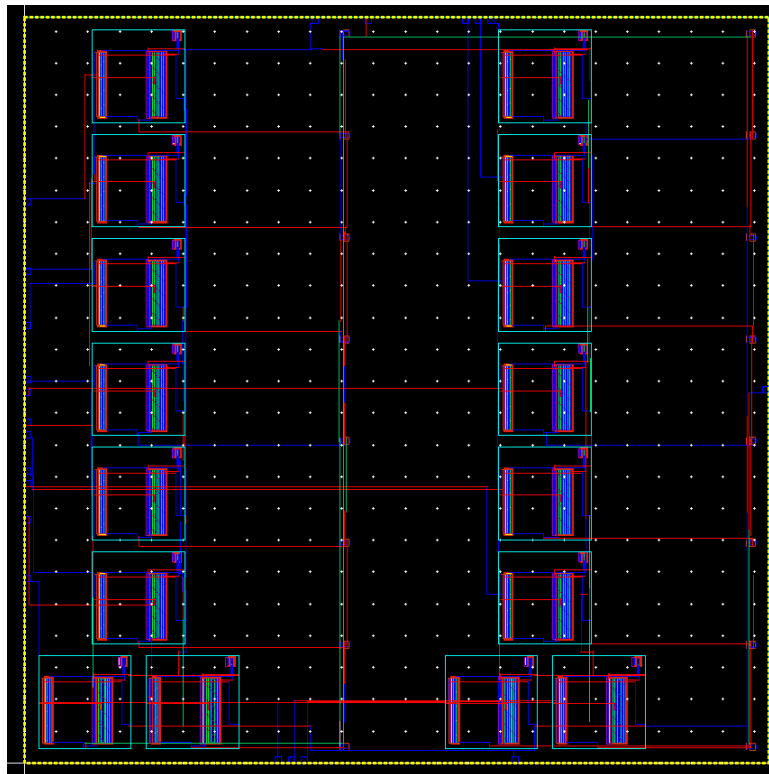


CMOS Transmission Gate DRC Pass

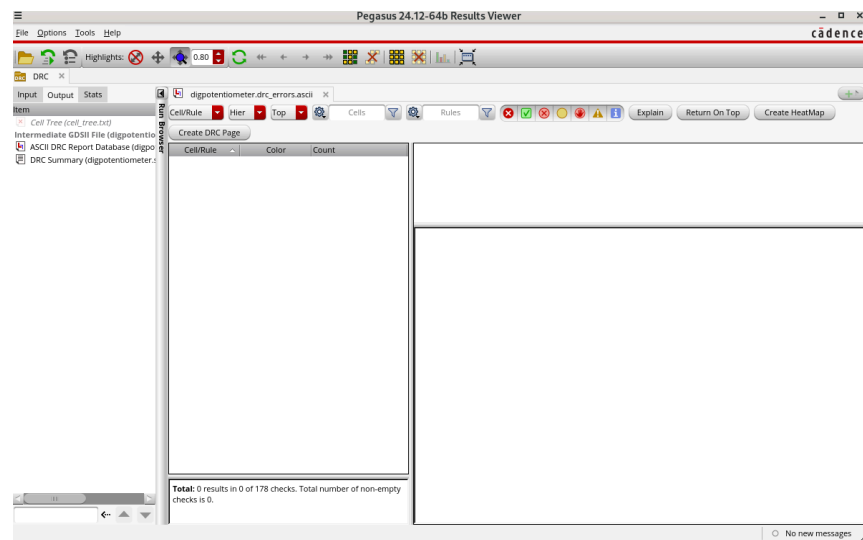


CMOS Transmission Gate LVS Pass

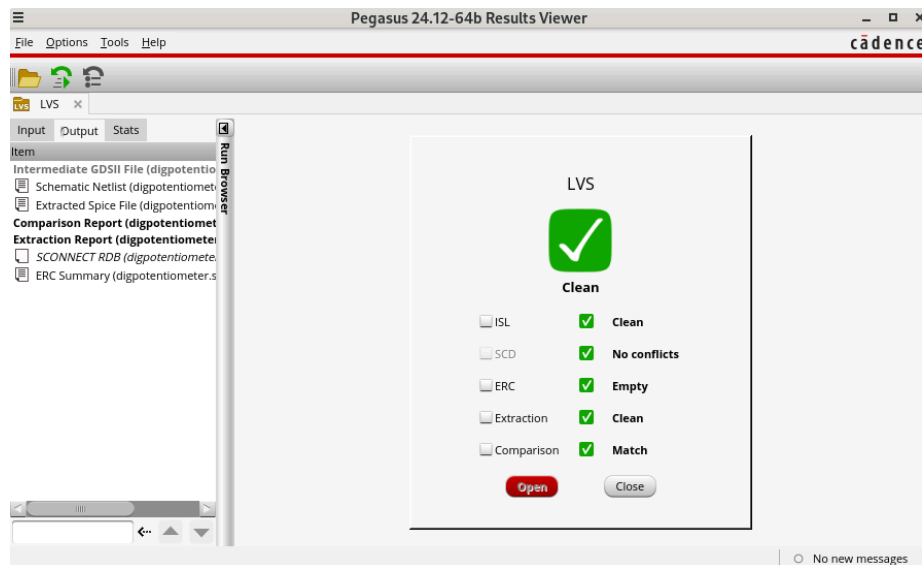
The next module requiring layout was the digital potentiometer cell. This was very straightforward, as it contained only 16 resistors and 16 transmission gates, allowing for an easy layout, which mostly worked with autorouting. After a few adjustments to pass DRC and LVS, this layout was finished. Note the resistors are three terminal devices, as they require a substrate connection. The resistors were polysilicon with silicide block resistors, allowing high and precise resistances. They were also relatively small, as can be seen.



Digital Potentiometer Layout



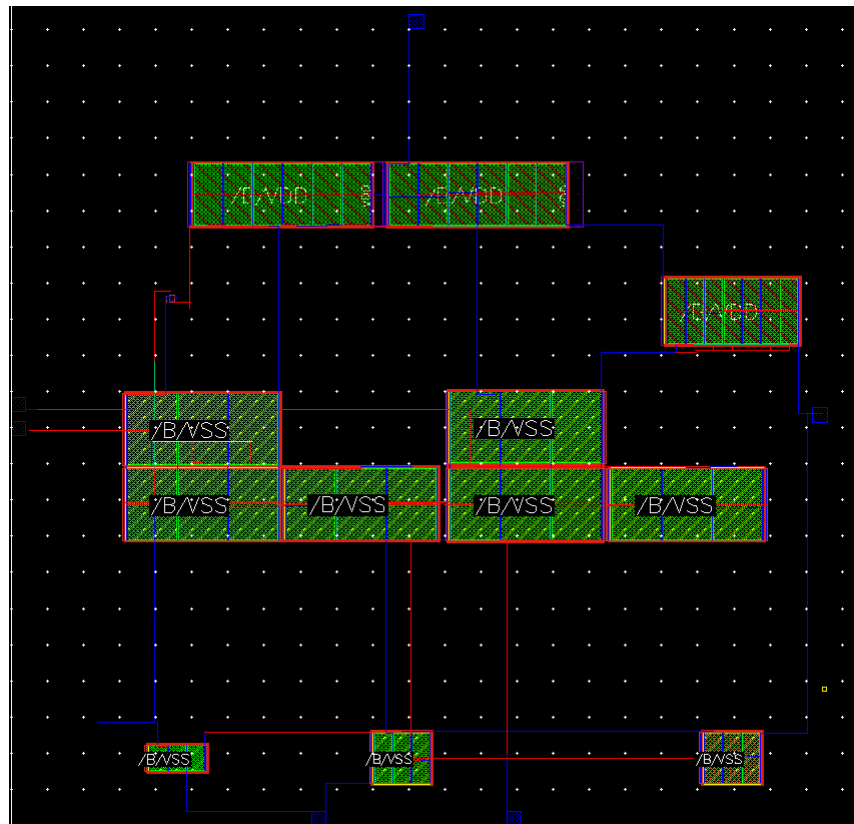
Digital Potentiometer DRC Pass



Digital Potentiometer LVS Pass

Moving to the op amp layout, using cell generation the first thing that was noticed was that the resistor was obscenely large (we initially used Metal 1 resistors). This wasn't an issue, per se, since there is no layout size requirement. That said, we still took the liberty to review the GPDK reference document and find another resistor that is more space conscious, while taking precautions to ensure this switch doesn't change the verification steps we took.

Looking at the layout, the use of fingers to preserve space and function within the GPDK045 structure, can be clearly seen in each of the transistors. This posed a challenge as we had very limited experience with the layout of several fingered devices up until this point. Lining up the inputs to several points on each device took some time to get used to. While the automatic cell generation gave a decent idea as to how they should be wired, it made several line up errors in DRC where vias in each transistor didn't have enough connection area with the layer above. This is where some guess and check work was done to fully grasp what the aims of the layout were.

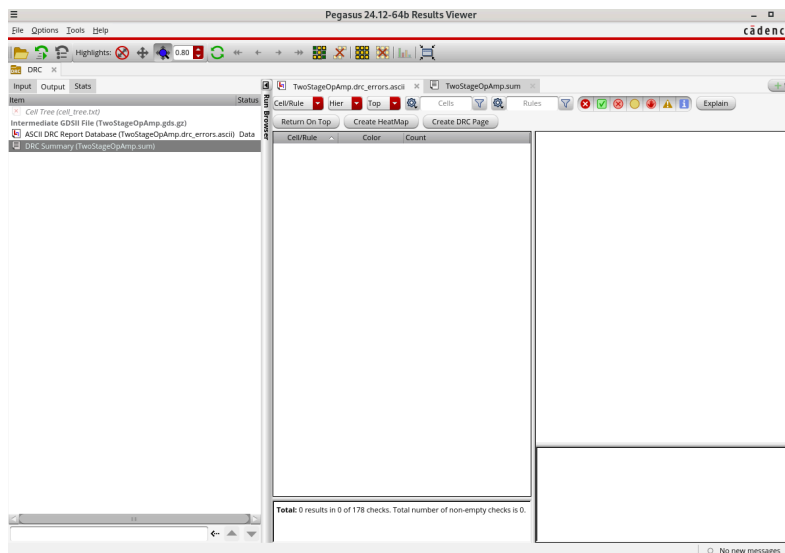


Op Amp Layout

Even with the resistor sizing refined, the layout is still rather unoptimized for space. Our team didn't give much attention towards making a compact design as it wasn't in our required project parameters. In a real world scenario, optimizing the transistor spacings would be a large part of this process. Another neglected aspect of physical layout was the application of common-centroid techniques. This approach would ensure that M1 and M2 or M3 and M4 experience the same manufacturing variations, which is necessary to get closer to achieving the target specifications for gain and stability that was simulated in our open loop test bench.

Running DRC, there were several layout errors that auto layout failed to consider. A few of the corrections that had to be made include repositioning the interconnect chains that connect each individual finger of each transistor for proper overlap, adding well taps to both sides of long

transistors to resolve proximity issues, and spacing between the transistors themselves. After significant layout refinement, the DRC successfully passed.



Op-Amp DRC Pass

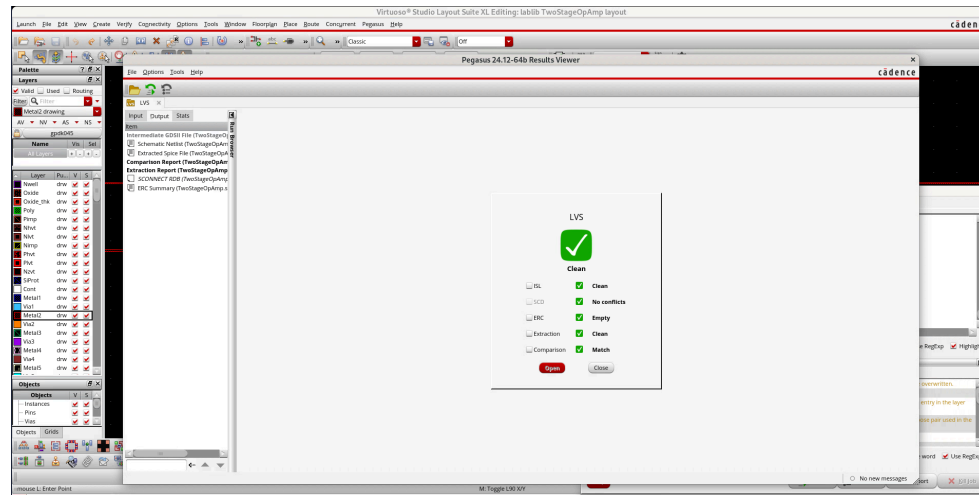
Group by Cell		Error Count	Status Det
TwoStageOpAmp		11	
MISMATCHED INSTANCES		2	
MISMATCHED INSTANCE PARAMETERS		9	
Merged Instance Detail			
mip1			
mip2			
mip3			
mip4			
mip5			
mip6			
mip7			
mip8			

Layout Instance	Schematic Instance	Parameter Error
<X190/M4> @(101.230,66.625) MP(g45p2svt) w: 62.615 u	MM0 MP(G45P2SVT) w: 62.6 u	(mip 5) 0.0239617%

Issues with LVS (Sizing reality vs ideal)

This error illuminated the fundamental conflict between continuous mathematical design and discrete physical manufacturing. The ideal width we calculated via Randy Geiger's sizing spreadsheet was exactly 62.6 μ m. However, the GPDK045 physical layout grid is constrained to strict fabrication machining sizings, specifically snapping to a resolution that includes 62.615 μ m to accommodate the physical limits of photolithography masks. When the parameterized cell was instantiated in the layout, the Cadence tool automatically forced the continuous 62.6 μ m width to

snap to the nearest valid manufacturing grid point, which was $62.615\mu\text{m}$. Since LVS demands exact, literal parametric matching, it flagged this necessary quantization as a fatal mismatch failure, even though it realistically would not cause any functional changes. To resolve this grid-snapping error, we back-annotated the schematic parameter to explicitly read $62.615\mu\text{m}$. Following this adjustment for each resistor, the Op-Amp successfully passed its final LVS check.

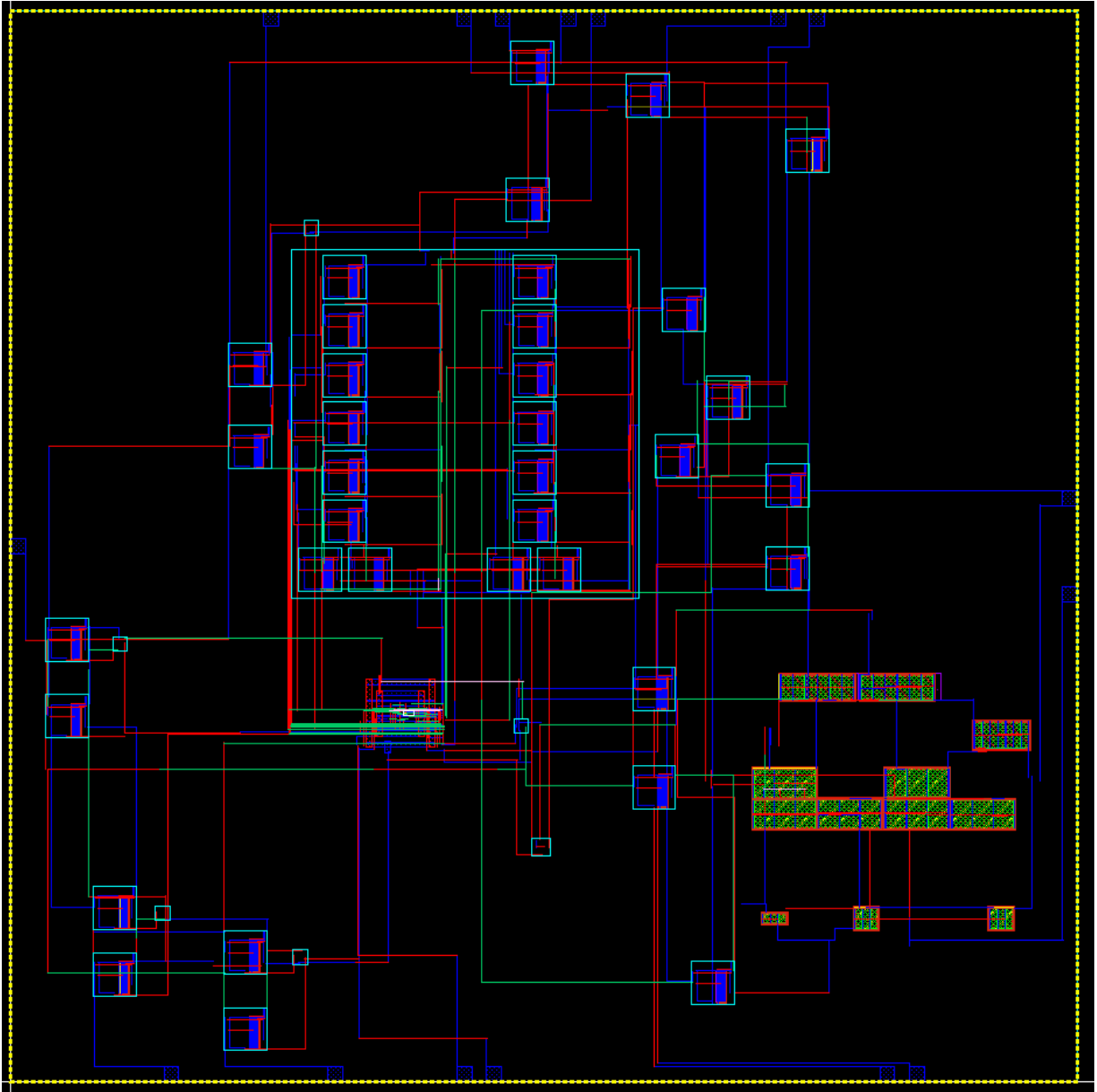


Op-Amp LVS Pass

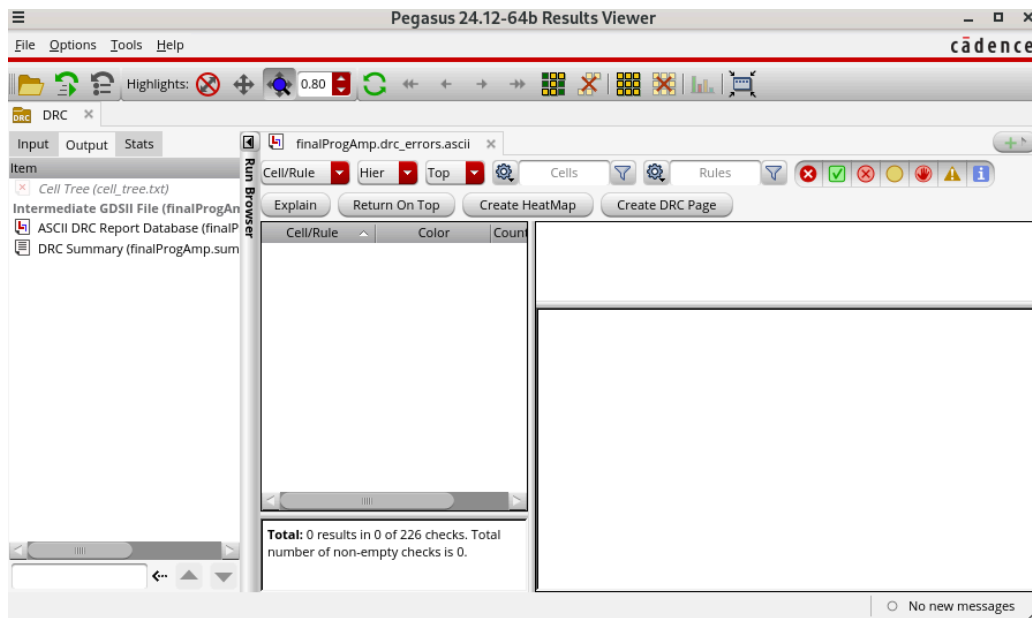
After completing the layout of all of the modules individually, the final layout process was fortunately very straightforward. Beginning with the schematic, it was very simple to use the “Generate All From Source” button to pull in all previous layout cellviews. From there, some rough floorplanning was done, in order to somewhat organize the cells by their connectivity and function. It can be seen that our final layout very closely resembles the schematic.

With the use of autorouting, 95% of the routing was quickly done entirely automatically, with just a few DRC issues requiring resolution. These issues were, as mentioned in previous layouts, simple issues regarding via spacing and other spacing issues occurring in adjacent metal traces. Within just a few minutes, these DRC issues were entirely resolved, meaning that the final step in completing our project was just passing LVS.

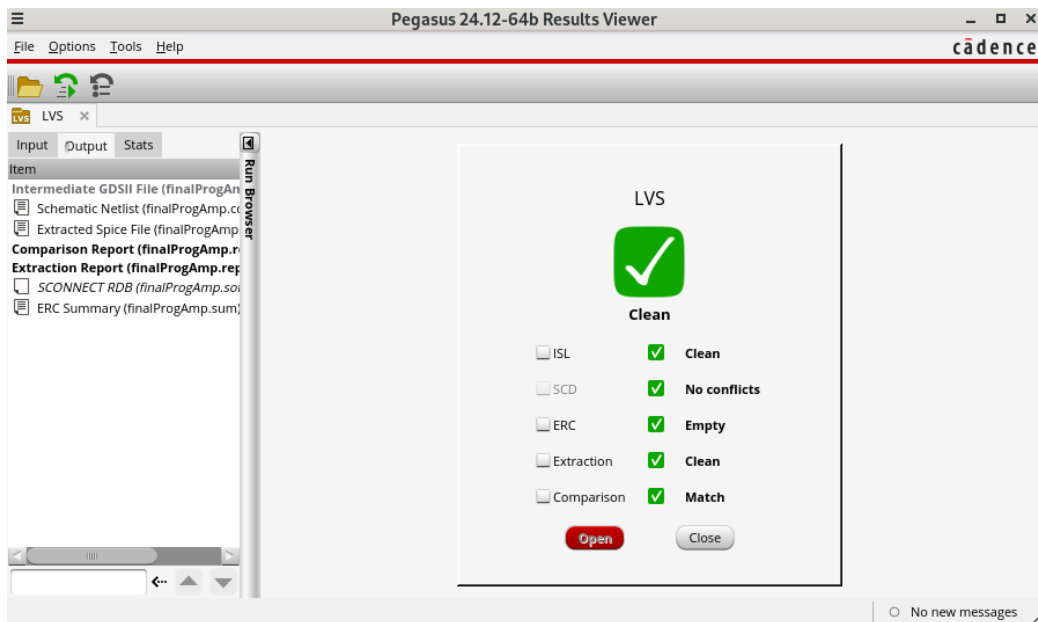
Our hopes were not too high going into LVS, as it seemed inevitable that something would be at least slightly wrong. In most of the previous layouts, a very simple issue was overlooked in the first draft of layout and was found in LVS. Fortunately, about a minute after starting the LVS, we were presented with an LVS pass on our first try. After wiping our tears of joy away, we began writing our report.



Full Layout (360.43um by 360.43um)



Full Layout DRC Pass



Full Layout LVS Pass

5. Conclusion and Project Reflection

This project succeeded in its aim of being a culmination of all the lab and coursework that we have covered in EE3300 in this semester. The design of this mixed-signal device utilized decoders, a 16-tap digital potentiometer, CMOS transmission gates, 7T operational amplifier, as well as other auxiliary logic components. This required thorough knowledge of both design tools (Cadence software) and the electrical components themselves: both things we managed to learn a lot about in this semester of EE330.

Designing analog components within a digital-centric PDK presented unique challenges we had not fully anticipated. The grid-snapping conflict between our ideal analog sizing and the discrete manufacturing constraints served as a stand-out learning experience. Another lesson was the resolution of the strong DAC signal ringing that we were getting in initial test benching required us to correct an analog system failure by physically up-sizing the geometry of our switches, decreasing their effective resistance. These specific experiences highlighted the unique approach one must take to mixed-signal design. It requires a sharp eye and a broad knowledge base to know how the device you are designing is integrated together so that possible solutions can be brainstormed once a check doesn't go the way that it was intended to.

In conclusion, our mixed-signal integrated circuit successfully demonstrated all four operation modes under transient simulation. Operating smoothly on the $\pm 1.8\text{V}$ supply, the device closely emulated the self-targeted programmable inverting gain limit of 15, the non-inverting gain limit of 16, a highly linear 16-tap 500-Ohm potentiometer response, and a precise 4-bit DAC output utilizing a 1V reference. Our physical verification was able to show that the entire integrated circuit meets all GPDK045 physical rules and functional specifications, resulting in a robust, if not fully size optimized implementation.

Works Cited

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